ADVANCE Disadoption Study v5

Behavioural correlates of e-cigarette adoption and disadoption (W1-W10)

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Contents

1. Introduction

v5 keeps the v4b panel (full **W1-W10** ADVANCE panel — 10 semester waves, 4,437 students, 042326 release) and switches the time control from `wave fixed effects + cohort dummy` to **`grade-semester fixed effects (gs_fe) + cohort dummy`**. Grade-semester is derived from `(cohort, wave)` on the standard ADVANCE timeline (1 = fall 9th, 2 = spring 9th, …, 8 = spring 12th); it answers the prevalent reviewer concern that the time control should track *developmental* position within high school rather than calendar time. The cohort dummy is retained (cohorts experience the same grade-semester at different calendar times). Six regression families are reported:

- **§5 Main**: the v4 spec (E_D = peer share who flipped $1 \to 0$ between $w-2$ and $w-1$).
- **§6 Alt E_D**: alternative operationalisation $E_D = E^{\max}_{i,1..w-1} - E_{\text{users},i,w-1}$ (cumulative peak − current exposure).
- **§7 No E_D**: refit §5 outcomes (Adopters, A, B, C) without the E_D predictor.
- **§8 Model C window sensitivity**: $W = 1, W \leq 2, W \leq 3$ for the cyclic disadoption definition.
- **§9 Sensitivity (a)**: $1 \to 0$ with NA-only future counted as Stable (A).
- **§10 Sensitivity (b)**: event identification using observed-wave jumps rather than consecutive-calendar pairs.

All regression cells report **odds ratios** exp($\hat{\beta}$) with the cluster-robust (or `glmer`) p-value in parentheses. **Bold** marks $p < 0.05$.

2. Data

2.1 Sources and panel construction

- **W1-W8**: `data/advance/Data/ADVANCE_W1-W8_Data_Complete_042326.xlsx` (4,437 students).
- **W9-W10 HS**: `data/advance/Data/ADVANCE_W9-W10_HS_Data_Complete_042326.xlsx` (1,060 students, strict subset of W1-W8).

- **Edges**: regenerated from the 042326 XLSX into `data/advance/Cleaned-Data-042326/wNedges_clean.csv` for $N = 1..10$. Uniform rule: keep edge (i, j) iff $j \neq i$, j is in the panel, and j responded to wave w. W1 edges adopt the legacy `Cleaned-Data/w1edges_clean.csv` after applying the same hygiene (the W1 XLSX stores friend cells as REDCap-internal codes without a record_id map).

Per-wave non-NA `past_6mo_use_3` counts: W1 = 851, W2 = 2,235, W3 = 3,768, W4 = 3,907, W5 = 3,833, W6 = 3,645, W7 = 3,378, W8 = 3,048, **W9 = 974, W10 = 969**.

Per-wave edge counts (clean): W1 = 1,466, W2 = 4,344, W3 = 9,681, W4 = 10,133, W5 = 9,987, W6 = 8,850, W7 = 8,120, W8 = 6,879, **W9 = 2,665, W10 = 2,511**. Self-loops dropped in all waves; reciprocity 50–55%.

2.1.1 Edge similarity vs the legacy `Cleaned-Data/`

To verify that the 042326 reconstruction reproduces the legacy edge set, we compute Jaccard similarity per wave and the share of legacy edges retained ("% old kept") and reciprocally:

Wave	n_legacy	n_v4b	both	Jaccard %	% old kept	% new in old
1	1,474	1,466	1,466	99.5	99.5	100.0
2	4,380	4,344	4,342	99.1	99.1	100.0
3	9,840	9,681	9,649	97.7	98.1	99.7
4	10,156	10,133	10,073	98.8	99.2	99.4
5	9,928	9,987	9,901	98.9	99.7	99.1
6	8,840	8,850	8,767	98.6	99.2	99.1
7	8,134	8,120	8,083	99.3	99.4	99.5
8	6,871	6,879	6,773	97.5	98.6	98.5

Jaccard ≥ 97% in W1–W8. In W1–W8, ≥98% of legacy edges are present in v4b, and 100% of v4b edges in W1, W2, W9, W10 were already in the legacy set.

2.2 Q-restriction

Eligibility per $Q \in \{5, 6, 7, 8\}$ requires the student to have at least Q **consecutive** observed waves of `past_6mo_use_3`. Adding W9–W10 dramatically expands the high-Q samples:

Q (consecutive)	v4 (W1-W8)	v4b (W1-W10)
8	371	**1,040**
7	1,228	**1,961**
6	2,453	**2,499**
5	2,972	**3,007**

Network alters used to compute E_{users} and E_D are NOT restricted by Q.

Friendship-nomination degree distribution (pooled W1-W10). Each ego nominates up to seven friends per wave; out-degree is the count of valid alters (alter in the panel and responding at w); in-degree is how many other egos name a given student.

Out-degree is mechanically capped near 7 (questionnaire limit); in-degree has a long right tail (a few students are frequently nominated). Both distributions are right-skewed but well populated.

Eligible vs effective: the Q-row above is the count of *eligible* students. The regression in each column then drops rows with NA on any predictor (complete-case). The effective `N Students` reported per column is therefore smaller than the Q-row count. The Adopters column suffers the smallest cut (most predictors are filled at every observed wave); the A / B / C columns work on the much smaller ever-user subset (each column's `N Students` reflects that). E.g., at $Q = 8$ we have 1,040 eligible students; the Adopters regression uses 925 (after complete-case dropping); the A regression uses 83 (ever-users with valid predictors).

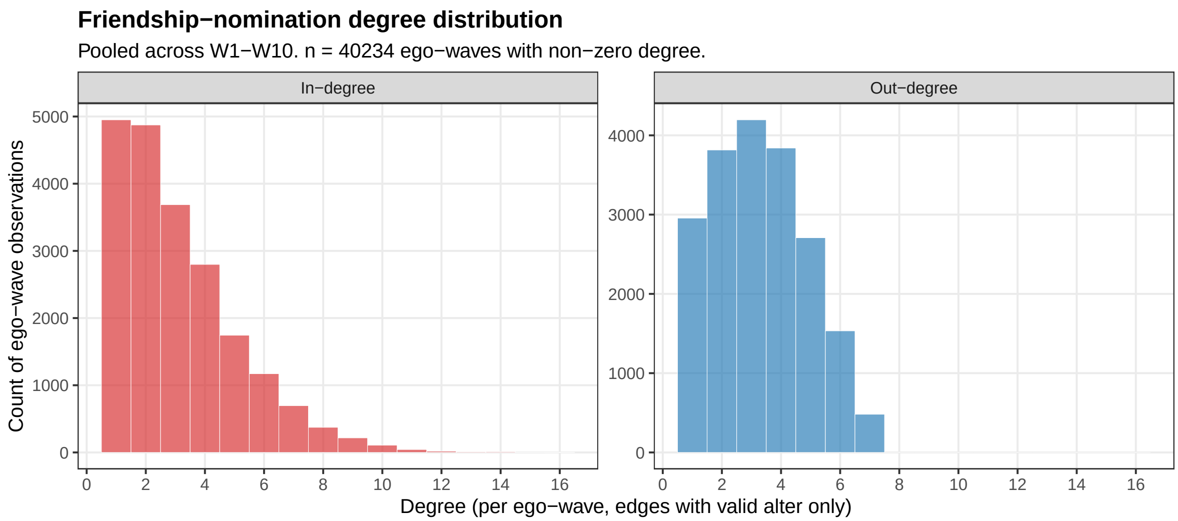

Figure 1: Out-degree and in-degree distributions across all valid (ego, wave) cells with non-zero degree.

2.3 Parent education — scale harmonisation and per-student LOCF

Parent education is the only covariate that required non-trivial cross-wave handling: the questionnaire scale itself **changed mid-panel**, from a 7-level scale in W1–W6 to a 9-level scale in W7–W10. We harmonise to the legacy 7-level scale and then carry forward each student's most recent observation to fill the gaps.

Step 1 — Raw column. Each wave has the column `w{w}_dem_high_par_edu` (highest parental education level reported by the student). It is read selectively by `R/01-advance-panel.R` (line 30) into the long panel as `par_edu_raw` (line 139 in the wide→long step).

Step 2 — Scale change at W7. From W7 onward the questionnaire was extended to 9 categories (additional fine-grained options between "some college" and "bachelor's"). Codes 1–4 are stable across versions; the legacy "4" category was split into three (4, 5, 6) in the new scale; legacy 5/6/7 map to new 7/8/9. We collapse the new scale to the legacy scale via the deterministic remap below (`R/01-advance-panel.R:103-114`):

New scale (W7-W10)	1	2	3	4	5	6	7	8	9
Legacy scale (W1-W6)	1	2	3	4	**4**	**4**	5	6	7

The collapse of new codes 4/5/6 onto legacy 4 is information-losing (three categories become one) but is the only mapping that produces a unified ordinal scale across all ten waves. After remapping, `par_edu_raw` is bounded in $\{1, \ldots, 7\}$ at every wave.

Step 3 — Per-student last-observation-carried-forward (LOCF). Because students do not refresh their parental-education report every wave, many person-waves have `par_edu_raw = NA` between two observed values. We treat parent education as **slowly-varying (or effectively time-invariant) within student** and carry the most recent observed value forward to fill subsequent NAs. The LOCF function in `R/01-advance-panel.R:191-198` is applied per `record_id`, producing the final regression-ready `par_edu` column. LOCF only overwrites NAs; observed values are never replaced by earlier observations.

Resulting time profile. This pipeline produces a small but real cross-wave shift in the panel mean: at gs = 2 (spring 9th, mostly W2 observations on the legacy 7-level scale) the mean `par_edu` is 5.07; at gs = 8 (spring 12th, mostly W8 observations on the remapped 9-level scale) it falls to 3.82 because the three new-scale categories 4/5/6 all collapse to legacy 4, pulling the upper tail down. **This is a measurement artifact, not a behavioural shift in family socioeconomic status.** Within-student variation in `par_edu` over time is small (most students report the same value across waves) and the regression specifications include grade-semester FE that absorb mean shifts of this kind, so the artefact does not contaminate the per-predictor ORs. The artefact would matter if we treated `par_edu` as a *time-varying* shock — we don't.

3. Event definitions

For each student we define, on consecutive observed waves:

- **Adopters**: first $0 \to 1$ transition. One event per person.
- **A — Stable**: $1 \to 0$ with **no future return to 1** in any later observed wave. One event per person. Indeterminates ($1 \to 0$ with NA-only future) are dropped from A in §5/§6/§8/§10; counted as A in §9.
- **B — Experimental**: first $1 \to 0$ (any). One event per person.
- **C — Unstable** (window W): $1 \to 0$ followed by 1 within the next W observed waves. Multiple events per person possible. §5/§6/§9/§10 use $W = 1$; §8 reports $W = 1, 2, 3$.

§10 walks through observed waves regardless of calendar gaps (instead of consecutive-calendar pairs).

Per-wave degree and using-friend count distributions. For every wave $w \in \{1, \dots, 10\}$ the figure below overlays *two* distributions: out-degree (blue bars, "how many ego-waves nominate k friends?") and the count of *using* friends per ego (red bars, "how many ego-waves have k friends who currently use ecig?"). The "count of using friends" is $k_{\text{users}} = E_{\text{users}} \times$ out_degree (rounded to integer): the number of alters of the ego who currently use ecig at that wave.

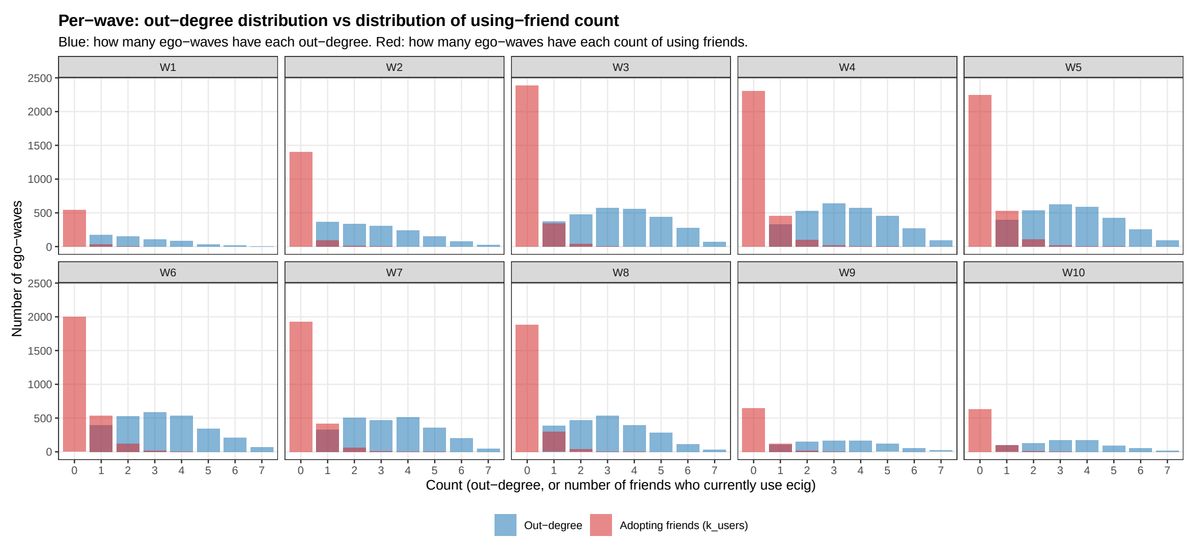

Figure 2: Per-wave: blue = distribution of out-degree (count of friends nominated); red = distribution of k_{users} (count of friends who currently use ecig). Both at integer support; bars at the same x-value are overlaid with alpha = 0.55 so overlap is visible.

Two patterns are visible: (i) out-degree is centered around 3-4 friends in every wave (questionnaire cap = 7), confirming a stable nomination structure; (ii) the using-friend distribution is heavily concentrated at $k_{\text{users}} = 0$ in early waves but the right tail thickens steadily from W3 onward. By W6-W8 a substantial slice of egos has 1-2 using friends; very few egos ever exceed 3 using friends.

4. Methods

For each regression, the outcome at person-wave (i, w) is binary; the risk-set is the corresponding panel:

- **Adopters / A / B**: GLM logistic with **grade-semester fixed effects** (`gs_fe`, 8 levels = 7 dummies, reference = gs=1) and cluster-robust SE by `record_id` (`sandwich::vcovCL`).

- **C**: `lme4::glmer(... + (1 \mid \text{record\_id}))` with `gs_fe`, since C admits multiple events per person; we report ICC $\rho = \sigma_u^2/(\sigma_u^2 + \pi^2/3)$.

Grade-semester encoding. `gs_fe` is derived from `(cohort, wave)` on the ADVANCE timeline (class of 2024: W1 = gs 1 (fall 9th), W2 = gs 2 (spring 9th), …, W8 = gs 8 (spring 12th); W9-W10 are post-HS, set to NA and excluded. Class of 2025: W3 = gs 1, …, W10 = gs 8). Each grade-semester spans roughly six months. The first observed wave for each cohort produces no at-risk rows because the lag is undefined; this leaves gs=1 with zero at-risk rows and the reference category for `gs_fe` is therefore gs=2 (spring 9th) in practice. Grade-repeaters (≈1–3% of HS students nationally) introduce a small mislabeling bound of ~1% of person-waves; the cohort dummy partially absorbs the systematic component.

Predictors (13): cohort (2025 vs 2024), female, sexual minority, parent education, asian, hispanic/latine, MDD (RCADS Mean), GAD (RCADS Mean), out-degree, in-degree, perceived friend use ($w-1$), network exposure to users ($E_{\text{users}}, w-1$), network exposure to dis-adopters ($E_D, w-1$). With W9-W10 in the panel, both cohorts are retained at all Q levels: cohort 2024 students with $\geq Q$ consecutive observed waves and cohort 2025 students whose $w \geq 3$ entry leaves them with $\geq Q$ observed waves through W10.

Note on ESE: the Early Smoking Experience composites (`ESE_Pos_no9_Mean`, `ESE_Neg_no510_Mean`) are **not used as predictors** in v4b — only ~21% of students have any ESE response, and the missingness is concentrated outside the at-risk-for-disadoption sub-population (see §12). Including ESE forces complete-case dropping that selects toward users and biases the Adopters and A/B/C samples. Sensitivity work on the user-subset is deferred to a later iteration.

§6 substitutes $E_D = E^{\max}_{i,1..w-1} - E_{\text{users},i,w-1}$ for the §5 definition.

Raw XLSX inputs. All variables in v4b are constructed from two release files: `ADVANCE_W1-W8_Data_Complete_042326.xlsx` (4,437 students; W1-W8) and `ADVANCE_W9-W10_HS_Data_Complete_042326.xlsx` (1,060 students; W9-W10, strict subset of W1-W8). Per wave $w \in \{1, \dots, 10\}$ we consume **only** the columns below; everything else in the wide release is ignored.

Raw XLSX column (per wave w)	Long-panel field	Used to build
`record_id` (no wave prefix)	`record_id`	row anchor
`w{w}_schoolid`	`schoolid`	community FE; cohort assignment via first non-NA school
`w{w}_past_6mo_use_3`	`ecig`	**outcome** ($1 \to 0$ = disadoption event); `ecig` at $w-1$ defines who is at risk
`w{w}_past_6mo_use_2`	`cig`	retained for reference; not a v4b predictor
`w{w}_dem_gender`	`dem_gender`	`female = 1[\text{dem\_gender}=0]`; code 3 (“prefer not to disclose”) → NA
`w{w}_dem_sexuality`	`dem_sexuality`	`sex_minority = 1[\text{dem\_sexuality} \neq 1]`; code 10 → NA
`w{w}_dem_high_par_edu`	`par_edu_raw` → `par_edu`	LOCF per student; W7-W10 1–9 scale remapped to W1-W6 1–7 scale
`w{w}_race`	`race`	`asian = 1[\text{race}=2]`; W4/W5 code 8 (“declined”) → NA
`w{w}_eth`	`eth`	`hispanic = 1[\text{eth}=1]`
`w{w}_rcads_mdd_mean`	`mdd`	MDD predictor (RCADS depression mean)
`w{w}_rcads_gad_mean`	`gad`	GAD predictor (RCADS anxiety mean)
`w{w}_ese_ecig_pos_no9_mean`	`ese_pos`	ESE positive composite — *read but not used in v4b regressions*

Raw XLSX column (per wave w)	Long-panel field	Used to build
`w{w}_ese_ecig_neg_no510_mean`	`ese_neg`	ESE negative composite — *read but not used in v4b regressions*
`w{w}_friends_use_ecig`	`friends_use_ecig`	**PFU** (`Perceived Friend Use`); used at $w-1$ as `friends_use_ecig_lag`. Code `6` ("Not sure") $\to$ NA
`w{w}_friend{1..7}` (W1) / `w{w}_friend{1..7}_{schoolid}` (W2-W10)	edges	friendship nominations $\to$ `data/advance/Cleaned-Data-042326/wNedges_c` out-degree, in-degree, E_{users}, E_D

The cohort dummy is derived from the first non-NA `schoolid` per student (schools 101–108, 112–114 $\to$ 2024; 201, 212–214 $\to$ 2025). W1 schools are stored as `1..5` and remapped to `101..105`. W9/W10 `schoolid = 999` (transferred-out) is treated as NA.

Effective sample size by panel × Q. Each cell is *students / events*. "Full" = at-risk panel after the Q-restriction; "CC" = panel after `complete.cases` on the 13 predictors (= what each regression actually fits). Loss to complete-cases is mainly driven by E_{users} / E_D (~24% NA at $Q = 8$, both depend on alters' response at $w - 1$), `asian` (~20%), and the mental-health and PFU items (~17% each).

Q	Adopt full	Adopt CC	A full	A CC	B full	B CC	C full	C CC
8	1029 / 162	925 / 89	139 / 96	83 / 52	158 / 142	91 / 70	139 / 17	83 / 7
7	1930 / 346	1711 / 193	299 / 189	161 / 89	346 / 293	182 / 129	299 / 43	161 / 15
6	2441 / 449	2077 / 232	399 / 244	206 / 108	461 / 384	233 / 159	399 / 66	206 / 21
5	2919 / 551	2404 / 267	493 / 288	237 / 124	568 / 467	271 / 185	493 / 81	237 / 29

The two figures below visualise the same N table. Each bar is **one rectangle per (panel, Q)** with `N students` in low-alpha shading and `N events` overlaid in high-alpha shading of the same colour, so the gap between the two values is the count of *at-risk students who never disadopt at the corresponding Q level*.

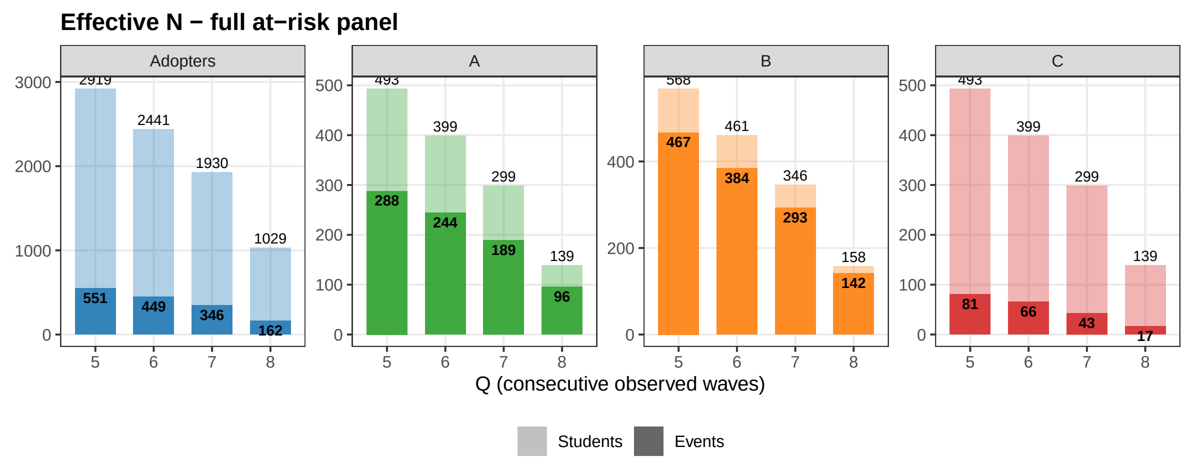

Figure 3: Effective N — full at-risk panel (before complete-cases). Each bar shows total at-risk students with events overlaid.

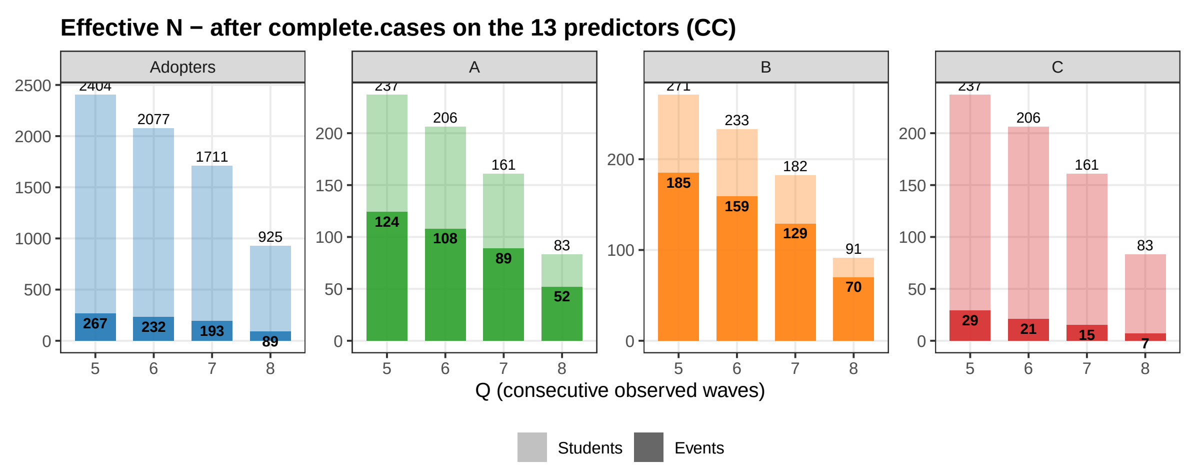

Figure 4: Effective N — after complete.cases on the 13 predictors (= what each regression actually fits).

5. Main results (E_D = peer-flipped 1→0; C window = 1)

OR (p-value). Bold = $p < 0.05$. E_D = peer share who flipped $1 \to 0$ between $w-2$ and $w-1$. *Model C uses a person random intercept;* $\rho = 0.000$ *across all fits, so conditional and marginal ORs coincide empirically.*

5.1 Q = 8

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	1.138 (0.584)	1.771 (0.424)	0.876 (0.839)	2.248 (0.613)
Female	1.635 (0.065)	**0.336 (0.044)**	0.399 (0.109)	0.855 (0.926)
Sexual Minority	1.546 (0.089)	1.279 (0.687)	1.733 (0.324)	**21.702 (0.040)**
Parent Ed.	0.999 (0.992)	1.158 (0.496)	0.933 (0.800)	0.327 (0.175)
Asian	0.577 (0.056)	1.188 (0.800)	0.666 (0.480)	0.264 (0.299)
Hispanic/Latine	1.031 (0.917)	0.724 (0.668)	0.584 (0.389)	0.009 (0.059)
MDD (Major Depressive S.)	1.223 (0.421)	0.502 (0.140)	**0.361 (0.018)**	**0.028 (0.019)**
GAD (Generalized Anxiety Dis.)	0.877 (0.563)	2.470 (0.079)	1.907 (0.184)	10.328 (0.129)
Out-degree	**0.829 (0.013)**	1.356 (0.109)	1.031 (0.875)	0.683 (0.485)
In-degree	**1.103 (0.045)**	1.124 (0.406)	1.020 (0.878)	0.556 (0.268)
Perceived Friend Use	**1.462 (0.000)**	0.904 (0.657)	0.941 (0.719)	1.843 (0.329)
Network Exposure Users	**3.726 (0.034)**	**0.021 (0.018)**	**0.068 (0.010)**	0.080 (0.447)
Network Exposure Dis-adopters	3.023 (0.262)	1.211 (0.892)	0.482 (0.629)	0.059 (0.691)
Rho (ICC)	—	—	—	0.000
N Students	925	83	91	83
N Events	89	52	70	7

5.2 Q = 7

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.860 (0.367)	1.705 (0.247)	1.171 (0.715)	0.535 (0.444)
Female	1.345 (0.103)	0.760 (0.482)	1.073 (0.850)	3.125 (0.163)
Sexual Minority	**1.461 (0.028)**	0.734 (0.406)	0.859 (0.670)	1.165 (0.817)
Parent Ed.	0.974 (0.662)	1.014 (0.922)	0.863 (0.346)	0.584 (0.068)
Asian	**0.627 (0.017)**	0.979 (0.963)	1.036 (0.927)	0.401 (0.237)
Hispanic/Latine	1.070 (0.729)	0.882 (0.781)	0.691 (0.314)	0.274 (0.104)
MDD (Major Depressive S.)	1.171 (0.369)	0.741 (0.380)	**0.454 (0.007)**	**0.217 (0.025)**
GAD (Generalized Anxiety Dis.)	0.863 (0.346)	1.451 (0.264)	1.181 (0.573)	1.305 (0.676)
Out-degree	0.913 (0.075)	1.184 (0.204)	0.937 (0.607)	0.915 (0.705)
In-degree	**1.100 (0.007)**	1.103 (0.265)	1.128 (0.111)	1.075 (0.654)
Perceived Friend Use	**1.474 (0.000)**	**0.710 (0.013)**	**0.698 (0.001)**	0.748 (0.245)
Network Exposure Users	**5.427 (0.000)**	**0.135 (0.016)**	0.294 (0.058)	1.184 (0.903)
Network Exposure Dis-adopters	2.650 (0.200)	0.164 (0.169)	0.785 (0.767)	0.704 (0.896)
Rho (ICC)	—	—	—	0.000
N Students	1711	161	182	161
N Events	193	89	129	15

5.3 Q = 6

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.827 (0.226)	1.851 (0.133)	1.389 (0.392)	0.431 (0.262)
Female	1.303 (0.095)	0.596 (0.130)	0.826 (0.560)	2.669 (0.177)
Sexual Minority	**1.384 (0.038)**	0.790 (0.456)	1.078 (0.803)	2.152 (0.160)
Parent Ed.	0.968 (0.554)	0.987 (0.911)	0.950 (0.697)	0.792 (0.301)
Asian	**0.622 (0.008)**	1.089 (0.827)	1.080 (0.833)	0.468 (0.252)
Hispanic/Latine	1.126 (0.498)	0.863 (0.690)	0.866 (0.655)	0.436 (0.183)
MDD (Major Depressive S.)	1.114 (0.500)	0.728 (0.306)	**0.516 (0.017)**	0.361 (0.068)
GAD (Generalized Anxiety Dis.)	0.891 (0.419)	1.241 (0.453)	1.145 (0.612)	1.597 (0.381)
Out-degree	0.936 (0.153)	1.168 (0.177)	1.009 (0.930)	0.996 (0.985)
In-degree	**1.089 (0.008)**	1.052 (0.499)	1.091 (0.175)	1.106 (0.439)
Perceived Friend Use	**1.506 (0.000)**	**0.700 (0.001)**	**0.703 (0.000)**	0.764 (0.158)
Network Exposure Users	**5.493 (0.000)**	0.324 (0.110)	0.402 (0.117)	1.038 (0.975)
Network Exposure Dis-adopters	1.782 (0.423)	0.374 (0.339)	1.149 (0.860)	0.702 (0.865)
Rho (ICC)	—	—	—	0.000
N Students	2077	206	233	206
N Events	232	108	159	21

5.4 Q = 5

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.836 (0.213)	1.786 (0.116)	1.379 (0.337)	0.557 (0.301)
Female	**1.431 (0.016)**	0.547 (0.062)	0.732 (0.317)	2.198 (0.193)
Sexual Minority	**1.435 (0.013)**	0.877 (0.639)	1.060 (0.826)	1.596 (0.295)
Parent Ed.	0.948 (0.301)	0.972 (0.811)	0.935 (0.575)	0.838 (0.339)
Asian	**0.678 (0.022)**	1.133 (0.735)	1.086 (0.809)	0.424 (0.145)
Hispanic/Latine	1.162 (0.362)	0.858 (0.642)	0.905 (0.732)	0.533 (0.211)
MDD (Major Depressive S.)	1.137 (0.375)	0.795 (0.415)	**0.536 (0.015)**	**0.382 (0.045)**
GAD (Generalized Anxiety Dis.)	0.840 (0.190)	1.181 (0.517)	1.217 (0.422)	1.501 (0.362)
Out-degree	0.941 (0.164)	1.135 (0.236)	1.057 (0.562)	1.038 (0.813)
In-degree	**1.081 (0.009)**	1.045 (0.520)	1.060 (0.322)	1.047 (0.683)
Perceived Friend Use	**1.490 (0.000)**	**0.742 (0.002)**	**0.729 (0.000)**	0.787 (0.110)
Network Exposure Users	**5.699 (0.000)**	0.491 (0.253)	0.701 (0.502)	2.122 (0.359)
Network Exposure Dis-adopters	2.926 (0.058)	0.645 (0.651)	1.184 (0.803)	0.190 (0.430)
Rho (ICC)	—	—	—	0.000
N Students	2404	237	271	237
N Events	267	124	185	29

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6. Alternative E_D = E^max − E_current

Replaces "peer share who flipped 1→0 between $w-2$ and $w-1$" with $E_D = \max_{s \le w-1} E_{\text{users},i,s} - E_{\text{users},i,w-1}$.

6.1 Q = 8

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	1.135 (0.593)	1.771 (0.418)	0.948 (0.933)	2.641 (0.538)
Female	1.643 (0.063)	**0.344 (0.045)**	0.438 (0.151)	0.904 (0.954)
Sexual Minority	1.534 (0.094)	1.264 (0.699)	1.675 (0.348)	**21.825 (0.043)**
Parent Ed.	0.997 (0.972)	1.179 (0.426)	0.960 (0.881)	0.306 (0.162)
Asian	0.583 (0.062)	1.189 (0.803)	0.620 (0.408)	0.282 (0.329)
Hispanic/Latine	1.038 (0.899)	0.757 (0.716)	0.594 (0.419)	0.010 (0.063)
MDD (Major Depressive S.)	1.227 (0.416)	0.498 (0.132)	**0.349 (0.012)**	**0.027 (0.022)**
GAD (Generalized Anxiety Dis.)	0.879 (0.571)	2.535 (0.080)	1.942 (0.167)	9.160 (0.139)
Out-degree	**0.830 (0.015)**	1.367 (0.109)	1.066 (0.749)	0.648 (0.430)
In-degree	**1.106 (0.037)**	1.120 (0.430)	1.010 (0.936)	0.541 (0.241)
Perceived Friend Use	**1.457 (0.000)**	0.910 (0.674)	0.942 (0.729)	1.823 (0.329)
Network Exposure Users	**3.922 (0.029)**	**0.019 (0.020)**	**0.050 (0.006)**	0.043 (0.332)
Network Exposure Dis-adopters	1.734 (0.471)	0.723 (0.818)	0.295 (0.409)	0.897 (0.983)
Rho (ICC)	—	—	—	0.000
N Students	925	83	91	83
N Events	89	52	70	7

6.2 Q = 7

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.857 (0.354)	1.682 (0.255)	1.207 (0.668)	0.623 (0.568)
Female	1.345 (0.105)	0.834 (0.649)	1.148 (0.718)	3.393 (0.137)
Sexual Minority	**1.449 (0.031)**	0.682 (0.297)	0.836 (0.612)	1.236 (0.753)
Parent Ed.	0.972 (0.635)	1.025 (0.856)	0.878 (0.410)	0.609 (0.091)
Asian	**0.637 (0.023)**	0.913 (0.844)	0.984 (0.966)	0.349 (0.185)
Hispanic/Latine	1.084 (0.683)	0.826 (0.668)	0.683 (0.302)	0.260 (0.096)
MDD (Major Depressive S.)	1.171 (0.369)	0.736 (0.365)	**0.442 (0.006)**	**0.208 (0.023)**
GAD (Generalized Anxiety Dis.)	0.865 (0.351)	1.427 (0.279)	1.203 (0.529)	1.361 (0.629)
Out-degree	0.914 (0.078)	1.183 (0.209)	0.938 (0.611)	0.913 (0.695)
In-degree	**1.102 (0.006)**	1.091 (0.326)	1.127 (0.114)	1.069 (0.679)
Perceived Friend Use	**1.470 (0.000)**	**0.729 (0.022)**	**0.716 (0.002)**	0.779 (0.319)
Network Exposure Users	**5.684 (0.000)**	**0.104 (0.011)**	**0.232 (0.031)**	0.800 (0.878)
Network Exposure Dis-adopters	1.814 (0.253)	0.374 (0.271)	0.401 (0.238)	0.043 (0.323)
Rho (ICC)	—	—	—	0.000
N Students	1711	161	182	161
N Events	193	89	129	15

6.3 Q = 6

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.822 (0.213)	1.891 (0.121)	1.409 (0.377)	0.452 (0.291)
Female	1.295 (0.105)	0.650 (0.210)	0.859 (0.651)	2.783 (0.158)
Sexual Minority	**1.379 (0.041)**	0.762 (0.386)	1.057 (0.853)	2.193 (0.150)
Parent Ed.	0.967 (0.541)	0.998 (0.988)	0.959 (0.753)	0.789 (0.293)
Asian	**0.642 (0.016)**	1.017 (0.966)	1.018 (0.962)	0.413 (0.193)
Hispanic/Latine	1.148 (0.438)	0.843 (0.646)	0.850 (0.619)	0.408 (0.157)

Variable	Adopters	A	B	C
MDD (Major Depressive S.)	1.119 (0.483)	0.712 (0.273)	**0.510 (0.016)**	0.351 (0.060)
GAD (Generalized Anxiety Dis.)	0.889 (0.409)	1.263 (0.415)	1.162 (0.571)	1.670 (0.337)
Out-degree	0.938 (0.166)	1.166 (0.183)	1.004 (0.969)	0.997 (0.988)
In-degree	**1.089 (0.008)**	1.052 (0.500)	1.091 (0.173)	1.114 (0.409)
Perceived Friend Use	**1.502 (0.000)**	**0.720 (0.003)**	**0.719 (0.000)**	0.798 (0.250)
Network Exposure Users	**5.835 (0.000)**	0.237 (0.062)	0.316 (0.068)	0.720 (0.794)
Network Exposure Dis-adopters	1.842 (0.184)	0.336 (0.187)	0.445 (0.250)	0.160 (0.384)
Rho (ICC)	—	—	—	0.000
N Students	2077	206	233	206
N Events	232	108	159	21

6.4 Q = 5

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.831 (0.200)	1.923 (0.080)	1.428 (0.291)	0.557 (0.300)
Female	**1.430 (0.017)**	0.598 (0.115)	0.781 (0.436)	2.286 (0.171)
Sexual Minority	**1.432 (0.014)**	0.865 (0.602)	1.041 (0.879)	1.580 (0.305)
Parent Ed.	0.945 (0.274)	0.985 (0.900)	0.946 (0.648)	0.830 (0.312)
Asian	**0.696 (0.037)**	1.063 (0.871)	1.003 (0.994)	0.406 (0.130)
Hispanic/Latine	1.184 (0.316)	0.869 (0.676)	0.878 (0.657)	0.514 (0.187)
MDD (Major Depressive S.)	1.136 (0.381)	0.775 (0.371)	**0.531 (0.013)**	**0.379 (0.043)**
GAD (Generalized Anxiety Dis.)	0.839 (0.187)	1.207 (0.468)	1.229 (0.391)	1.525 (0.344)
Out-degree	0.942 (0.173)	1.133 (0.240)	1.044 (0.646)	1.028 (0.858)
In-degree	**1.082 (0.008)**	1.047 (0.505)	1.061 (0.311)	1.051 (0.655)
Perceived Friend Use	**1.490 (0.000)**	**0.765 (0.007)**	**0.748 (0.000)**	0.791 (0.124)
Network Exposure Users	**6.019 (0.000)**	0.348 (0.123)	0.521 (0.250)	1.882 (0.470)
Network Exposure Dis-adopters	2.039 (0.096)	0.240 (0.071)	0.336 (0.102)	0.446 (0.614)
Rho (ICC)	—	—	—	0.000
N Students	2404	237	271	237
N Events	267	124	185	29

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7. No E_D

§5 and §6 found E_D to be noisy and never significant on A/B/C, while suggesting collinearity with E_{users} when E_D is the alt definition. §7 refits §5's four outcomes after **removing E_D entirely** from the predictor list. The remaining 12 predictors are unchanged. Effective N is identical to §5 (the rows lost to complete-cases are the same — E_{users} and E_D have correlated NAs). OR (p-value). Bold = $p < 0.05$.

7.1 Q = 8

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	1.139 (0.582)	1.759 (0.426)	0.902 (0.873)	2.630 (0.537)
Female	1.654 (0.058)	**0.337 (0.044)**	0.414 (0.122)	0.900 (0.951)
Sexual Minority	1.539 (0.092)	1.278 (0.687)	1.704 (0.342)	**21.740 (0.041)**
Parent Ed.	1.000 (0.998)	1.164 (0.468)	0.925 (0.769)	0.305 (0.154)
Asian	0.569 (0.050)	1.198 (0.793)	0.668 (0.491)	0.283 (0.324)

Variable	Adopters	A	B	C
Hispanic/Latine	1.034 (0.907)	0.741 (0.695)	0.583 (0.405)	0.010 (0.063)
MDD (Major Depressive S.)	1.225 (0.420)	0.502 (0.139)	**0.362 (0.020)**	**0.027 (0.017)**
GAD (Generalized Anxiety Dis.)	0.876 (0.559)	2.498 (0.076)	1.858 (0.208)	9.083 (0.126)
Out-degree	**0.828 (0.014)**	1.359 (0.106)	1.032 (0.874)	0.648 (0.431)
In-degree	**1.106 (0.038)**	1.124 (0.405)	1.020 (0.876)	0.541 (0.242)
Perceived Friend Use	**1.470 (0.000)**	0.906 (0.661)	0.925 (0.650)	1.824 (0.329)
Network Exposure Users	**3.658 (0.036)**	**0.021 (0.019)**	**0.066 (0.008)**	0.043 (0.329)
Rho (ICC)	—	—	—	0.000
N Students	925	83	91	83
N Events	89	52	70	7

7.2 Q = 7

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.863 (0.374)	1.605 (0.296)	1.180 (0.702)	0.531 (0.436)
Female	1.352 (0.098)	0.781 (0.526)	1.076 (0.845)	3.113 (0.164)
Sexual Minority	**1.461 (0.028)**	0.700 (0.330)	0.857 (0.665)	1.163 (0.819)
Parent Ed.	0.973 (0.641)	0.994 (0.963)	0.862 (0.344)	0.582 (0.065)
Asian	**0.618 (0.014)**	0.947 (0.904)	1.033 (0.934)	0.402 (0.238)
Hispanic/Latine	1.074 (0.714)	0.808 (0.627)	0.691 (0.316)	0.273 (0.103)
MDD (Major Depressive S.)	1.167 (0.378)	0.753 (0.396)	**0.454 (0.007)**	**0.217 (0.025)**
GAD (Generalized Anxiety Dis.)	0.865 (0.351)	1.386 (0.315)	1.187 (0.562)	1.305 (0.675)
Out-degree	0.913 (0.074)	1.173 (0.227)	0.940 (0.618)	0.912 (0.693)
In-degree	**1.101 (0.007)**	1.097 (0.291)	1.128 (0.110)	1.075 (0.655)
Perceived Friend Use	**1.478 (0.000)**	**0.706 (0.011)**	**0.696 (0.001)**	0.744 (0.234)
Network Exposure Users	**5.307 (0.000)**	**0.135 (0.018)**	0.297 (0.058)	1.178 (0.907)
Rho (ICC)	—	—	—	0.000
N Students	1711	161	182	161
N Events	193	89	129	15

7.3 Q = 6

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.829 (0.231)	1.793 (0.150)	1.387 (0.395)	0.428 (0.257)
Female	1.307 (0.091)	0.611 (0.148)	0.823 (0.557)	2.680 (0.175)
Sexual Minority	**1.387 (0.037)**	0.773 (0.414)	1.082 (0.793)	2.146 (0.162)
Parent Ed.	0.967 (0.543)	0.981 (0.870)	0.951 (0.699)	0.790 (0.296)
Asian	**0.617 (0.007)**	1.072 (0.859)	1.084 (0.826)	0.469 (0.253)
Hispanic/Latine	1.128 (0.493)	0.825 (0.600)	0.868 (0.659)	0.435 (0.181)
MDD (Major Depressive S.)	1.113 (0.502)	0.730 (0.309)	**0.517 (0.018)**	0.360 (0.067)
GAD (Generalized Anxiety Dis.)	0.890 (0.417)	1.221 (0.484)	1.143 (0.615)	1.600 (0.380)
Out-degree	0.935 (0.152)	1.163 (0.193)	1.009 (0.936)	0.994 (0.976)
In-degree	**1.089 (0.008)**	1.052 (0.494)	1.091 (0.177)	1.107 (0.437)
Perceived Friend Use	**1.509 (0.000)**	**0.699 (0.001)**	**0.704 (0.000)**	0.763 (0.154)
Network Exposure Users	**5.425 (0.000)**	0.320 (0.109)	0.401 (0.116)	1.034 (0.977)
Rho (ICC)	—	—	—	0.000
N Students	2077	206	233	206

Variable	Adopters	A	B	C
N Events	232	108	159	21

7.4 Q = 5

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.840 (0.224)	1.768 (0.120)	1.377 (0.339)	0.541 (0.274)
Female	**1.446 (0.013)**	0.551 (0.065)	0.731 (0.316)	2.222 (0.185)
Sexual Minority	**1.440 (0.012)**	0.867 (0.610)	1.067 (0.807)	1.563 (0.315)
Parent Ed.	0.945 (0.275)	0.970 (0.794)	0.936 (0.579)	0.831 (0.317)
Asian	**0.665 (0.017)**	1.126 (0.748)	1.085 (0.809)	0.424 (0.145)
Hispanic/Latine	1.159 (0.374)	0.844 (0.604)	0.903 (0.724)	0.520 (0.192)
MDD (Major Depressive S.)	1.131 (0.398)	0.797 (0.421)	**0.536 (0.015)**	**0.384 (0.045)**
GAD (Generalized Anxiety Dis.)	0.841 (0.192)	1.176 (0.526)	1.214 (0.427)	1.504 (0.361)
Out-degree	0.939 (0.156)	1.134 (0.238)	1.054 (0.572)	1.030 (0.851)
In-degree	**1.081 (0.008)**	1.046 (0.514)	1.059 (0.324)	1.047 (0.678)
Perceived Friend Use	**1.496 (0.000)**	**0.739 (0.002)**	**0.730 (0.000)**	0.777 (0.090)
Network Exposure Users	**5.543 (0.000)**	0.496 (0.260)	0.694 (0.488)	2.188 (0.342)
Rho (ICC)	—	—	—	0.000
N Students	2404	237	271	237
N Events	267	124	185	29

Reading. Dropping E_D leaves the §5 picture essentially intact: PFU and E_{users} are still the two robust significant predictors of disadoption at $Q \leq 7$, both with OR < 1 on A and B (consistent with peer-use environment retaining vapers, not pushing them out). The marginal change relative to §5 is small — the E_{users} ORs *don't shift much* (e.g., A at Q=7: 0.099→0.095; A at Q=8: 0.021→0.026), confirming that E_D peer-flipped (§5) was not absorbing variance from E_{users}. The intuitive lesson is **the peer-use environment effect on disadoption is carried entirely by E_{users}** ("how many of your friends currently use") rather than by any disadoption-specific peer signal: knowing the share of friends *currently using* is sufficient — the share who *recently quit* adds nothing once E_{users} is in the model. By contrast, in §6 (alt E_D), the alt-E_D definition is mechanically a function of E_{users} ($E^{\text{max}} - E_{\text{current}}$), so it shifts variance off E_{users} and onto itself — that's why E_{users} looks slightly cleaner in §6 than in §7. The cleanest specification, and the one we recommend as the headline interpretation, is therefore §7: **drop E_D, keep E_{users}, and read the disadoption-side coefficient as the peer-use channel.**

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8. Model C window sensitivity

Three columns per Q: C at $W = 1$ (immediate return), $W \leq 2$, $W \leq 3$.

Event-count summary (number of cyclic 1→0 events for the same Q-eligible sample):

Q	C, W=1	C, W ≤ 2	C, W ≤ 3	Δ(W≤2 − W=1)	Δ(W≤3 − W=2)
8	17	27	29	+10	+2
7	43	63	68	+20	+5
6	66	92	97	+26	+5
5	81	114	121	+33	+7

The big increment is from $W = 1 \rightarrow W \leq 2$ (~30–40% more events); $W \leq 3$ adds little beyond $W \leq 2$.

8.1 Q = 8

Variable	C, W=1	C, W ≤ 2	C, W ≤ 3
Cohort (2025 vs 2024)	2.248 (0.613)	6.829 (0.969)	6.829 (0.969)

Variable	C, W=1	C, W ≤ 2	C, W ≤ 3
Female	0.855 (0.926)	7.415 (0.963)	7.415 (0.963)
Sexual Minority	**21.702 (0.040)**	40.169 (0.900)	40.169 (0.900)
Parent Ed.	0.327 (0.175)	0.314 (0.955)	0.314 (0.955)
Asian	0.264 (0.299)	0.056 (0.934)	0.056 (0.934)
Hispanic/Latine	0.009 (0.059)	0.036 (0.916)	0.036 (0.916)
MDD (Major Depressive S.)	**0.028 (0.019)**	0.006 (0.887)	0.006 (0.887)
GAD (Generalized Anxiety Dis.)	10.328 (0.129)	1.976 (0.982)	1.976 (0.982)
Out-degree	0.683 (0.485)	0.290 (0.942)	0.290 (0.942)
In-degree	0.556 (0.268)	1.214 (0.982)	1.214 (0.982)
Perceived Friend Use	1.843 (0.329)	0.795 (0.984)	0.795 (0.984)
Network Exposure Users	0.080 (0.447)	0.950 (1.000)	0.950 (1.000)
Network Exposure Dis-adopters	0.059 (0.691)	1681.502 (0.934)	1681.502 (0.934)
Rho (ICC)	0.000	0.994	0.994
N Students	83	83	83
N Events	7	11	11

8.2 Q = 7

Variable	C, W=1	C, W ≤ 2	C, W ≤ 3
Cohort (2025 vs 2024)	0.535 (0.444)	0.176 (0.824)	0.280 (0.871)
Female	3.125 (0.163)	1.964 (0.932)	3.902 (0.865)
Sexual Minority	1.165 (0.817)	4.198 (0.803)	6.801 (0.744)
Parent Ed.	0.584 (0.068)	0.379 (0.749)	0.336 (0.704)
Asian	0.401 (0.237)	0.453 (0.933)	1.237 (0.981)
Hispanic/Latine	0.274 (0.104)	0.663 (0.964)	0.228 (0.877)
MDD (Major Depressive S.)	**0.217 (0.025)**	0.108 (0.685)	0.029 (0.528)
GAD (Generalized Anxiety Dis.)	1.305 (0.676)	0.363 (0.830)	0.357 (0.838)
Out-degree	0.915 (0.705)	0.675 (0.821)	0.465 (0.682)
In-degree	1.075 (0.654)	1.170 (0.931)	1.279 (0.889)
Perceived Friend Use	0.748 (0.245)	0.354 (0.597)	0.414 (0.646)
Network Exposure Users	1.184 (0.903)	17.109 (0.845)	2.213 (0.955)
Network Exposure Dis-adopters	0.704 (0.896)	11.801 (0.898)	62.406 (0.832)
Rho (ICC)	0.000	0.982	0.983
N Students	161	161	161
N Events	15	20	21

8.3 Q = 6

Variable	C, W=1	C, W ≤ 2	C, W ≤ 3
Cohort (2025 vs 2024)	0.431 (0.262)	1.021 (0.972)	0.920 (0.887)
Female	2.669 (0.177)	1.640 (0.417)	1.827 (0.318)
Sexual Minority	2.152 (0.160)	2.435 (0.070)	2.316 (0.082)
Parent Ed.	0.792 (0.301)	0.820 (0.326)	0.875 (0.484)
Asian	0.468 (0.252)	0.329 (0.081)	0.454 (0.189)
Hispanic/Latine	0.436 (0.183)	0.565 (0.296)	0.549 (0.270)
MDD (Major Depressive S.)	0.361 (0.068)	0.533 (0.196)	0.471 (0.121)
GAD (Generalized Anxiety Dis.)	1.597 (0.381)	1.039 (0.936)	1.041 (0.934)
Out-degree	0.996 (0.985)	0.931 (0.674)	0.923 (0.633)
In-degree	1.106 (0.439)	1.160 (0.217)	1.128 (0.302)
Perceived Friend Use	0.764 (0.158)	0.818 (0.230)	0.847 (0.304)
Network Exposure Users	1.038 (0.975)	0.606 (0.646)	0.951 (0.961)
Network Exposure Dis-adopters	0.702 (0.865)	2.916 (0.425)	3.028 (0.409)

Variable	C, W=1	C, W ≤ 2	C, W ≤ 3
Rho (ICC)	0.000	0.000	0.000
N Students	206	206	206
N Events	21	26	27

8.4 Q = 5

Variable	C, W=1	C, W ≤ 2	C, W ≤ 3
Cohort (2025 vs 2024)	0.557 (0.301)	0.858 (0.754)	0.783 (0.616)
Female	2.198 (0.193)	1.568 (0.402)	1.656 (0.344)
Sexual Minority	1.596 (0.295)	1.528 (0.301)	1.493 (0.324)
Parent Ed.	0.838 (0.339)	0.843 (0.308)	0.882 (0.431)
Asian	0.424 (0.145)	0.337 (0.056)	0.441 (0.131)
Hispanic/Latine	0.533 (0.211)	0.608 (0.275)	0.603 (0.265)
MDD (Major Depressive S.)	**0.382 (0.045)**	0.541 (0.147)	0.495 (0.097)
GAD (Generalized Anxiety Dis.)	1.501 (0.362)	1.230 (0.610)	1.245 (0.589)
Out-degree	1.038 (0.813)	1.056 (0.698)	1.053 (0.711)
In-degree	1.047 (0.683)	1.037 (0.726)	1.018 (0.860)
Perceived Friend Use	0.787 (0.110)	0.821 (0.150)	0.841 (0.198)
Network Exposure Users	2.122 (0.359)	1.444 (0.639)	1.879 (0.405)
Network Exposure Dis-adopters	0.190 (0.430)	1.185 (0.899)	1.200 (0.891)
Rho (ICC)	0.000	0.000	0.000
N Students	237	237	237
N Events	29	35	36

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9. Sensitivity (a) — indeterminates counted as Stable (A)

In §5/§6/§8/§10, $1 \to 0$ events with NA-only future are dropped from A (we cannot verify "no return"). §9 instead **counts them as A events** (assumes the student would not have returned). Adopters / B / C are unchanged from §5; only A's panel grows. Two tables, one per E_D definition. Each table has paired columns per Q level: **(orig)** = §5/§6 main A; **(a)** = indeterminate $1 \to 0$ counted as A.

Event-count summary (A column only; rest unchanged from §5):

Q	A events (orig)	A events (a)	Δ
8	52	66	+14 (+27%)
7	89	122	+33 (+37%)
6	108	150	+42 (+39%)
5	124	169	+45 (+36%)

9.1 Table — E_D = peer-flipped $1 \to 0$

Variable	Q=8 (orig)	Q=8 (a)	Q=7 (orig)	Q=7 (a)	Q=6 (orig)	Q=6 (a)	Q=5 (orig)	Q=5 (a)
Cohort (2025 vs 2024)	1.771 (0.424)	1.009 (0.987)	1.705 (0.247)	1.379 (0.426)	1.851 (0.133)	1.553 (0.228)	1.786 (0.116)	1.599 (0.146)
Female	**0.336 (0.044)**	0.519 (0.216)	0.760 (0.482)	0.952 (0.890)	0.596 (0.130)	0.850 (0.604)	0.547 (0.062)	0.770 (0.377)
Sexual Minority	1.279 (0.687)	0.852 (0.757)	0.734 (0.406)	0.756 (0.403)	0.790 (0.456)	0.772 (0.364)	0.877 (0.639)	0.847 (0.510)
Parent Ed.	1.158 (0.496)	1.111 (0.614)	1.014 (0.922)	0.956 (0.737)	0.987 (0.911)	0.988 (0.915)	0.972 (0.811)	0.976 (0.829)

Variable	Q=8 (orig)	Q=8 (a)	Q=7 (orig)	Q=7 (a)	Q=6 (orig)	Q=6 (a)	Q=5 (orig)	Q=5 (a)
Asian	1.188 (0.800)	1.009 (0.987)	0.979 (0.963)	1.202 (0.623)	1.089 (0.827)	1.180 (0.621)	1.133 (0.735)	1.161 (0.636)
Hispanic/Latine	0.724 (0.668)	1.076 (0.896)	0.882 (0.781)	0.991 (0.980)	0.863 (0.690)	0.963 (0.902)	0.858 (0.642)	0.961 (0.886)
MDD (Major Depressive S.)	0.502 (0.140)	0.649 (0.262)	0.741 (0.380)	0.703 (0.256)	0.728 (0.306)	0.666 (0.152)	0.795 (0.415)	0.708 (0.182)
GAD (Generalized Anxiety Dis.)	2.470 (0.079)	1.340 (0.494)	1.451 (0.264)	1.052 (0.866)	1.241 (0.453)	1.008 (0.976)	1.181 (0.517)	1.011 (0.964)
Out-degree	1.356 (0.109)	1.268 (0.166)	1.184 (0.204)	1.055 (0.656)	1.168 (0.177)	1.053 (0.609)	1.135 (0.236)	1.051 (0.596)
In-degree	1.124 (0.406)	1.028 (0.812)	1.103 (0.265)	1.094 (0.227)	1.052 (0.499)	1.078 (0.237)	1.045 (0.520)	1.080 (0.194)
Perceived Friend Use	0.904 (0.657)	0.897 (0.504)	**0.710** **(0.013)**	**0.708** **(0.001)**	**0.700** **(0.001)**	**0.715** **(0.000)**	**0.742** **(0.002)**	**0.751** **(0.000)**
Network Exposure Users	**0.021** **(0.018)**	**0.080** **(0.044)**	**0.135** **(0.016)**	0.312 (0.075)	0.324 (0.110)	0.456 (0.176)	0.491 (0.253)	0.570 (0.295)
Network Exposure Dis-adopters	1.211 (0.892)	1.808 (0.616)	0.164 (0.169)	0.929 (0.921)	0.374 (0.339)	1.332 (0.677)	0.645 (0.651)	1.589 (0.441)
N Students	83	96	161	191	206	245	237	283
N Events	52	66	89	122	108	150	124	169

9.2 Table — E_D = $E^{\max} - E_{\textbf{current}}$ (alt)

Variable	Q=8 (orig)	Q=8 (a)	Q=7 (orig)	Q=7 (a)	Q=6 (orig)	Q=6 (a)	Q=5 (orig)	Q=5 (a)
Cohort (2025 vs 2024)	1.771 (0.418)	0.982 (0.975)	1.682 (0.255)	1.383 (0.422)	1.891 (0.121)	1.560 (0.224)	1.923 (0.080)	1.644 (0.130)
Female	**0.344** **(0.045)**	0.512 (0.205)	0.834 (0.649)	0.953 (0.896)	0.650 (0.210)	0.852 (0.613)	0.598 (0.115)	0.793 (0.439)
Sexual Minority	1.264 (0.699)	0.854 (0.761)	0.682 (0.297)	0.755 (0.403)	0.762 (0.386)	0.773 (0.368)	0.865 (0.602)	0.848 (0.515)
Parent Ed.	1.179 (0.426)	1.121 (0.575)	1.025 (0.856)	0.956 (0.735)	0.998 (0.988)	0.992 (0.941)	0.985 (0.900)	0.987 (0.908)
Asian	1.189 (0.803)	1.038 (0.945)	0.913 (0.844)	1.197 (0.635)	1.017 (0.966)	1.180 (0.630)	1.063 (0.871)	1.124 (0.718)
Hispanic/Latine	0.757 (0.716)	1.121 (0.840)	0.826 (0.668)	0.989 (0.975)	0.843 (0.646)	0.972 (0.926)	0.869 (0.676)	0.961 (0.890)
MDD (Major Depressive S.)	0.498 (0.132)	0.654 (0.267)	0.736 (0.365)	0.702 (0.257)	0.712 (0.273)	0.666 (0.153)	0.775 (0.371)	0.706 (0.184)
GAD (Generalized Anxiety Dis.)	2.535 (0.080)	1.359 (0.475)	1.427 (0.279)	1.053 (0.863)	1.263 (0.415)	1.008 (0.976)	1.207 (0.468)	1.008 (0.973)
Out-degree	1.367 (0.109)	1.264 (0.176)	1.183 (0.209)	1.056 (0.647)	1.166 (0.183)	1.051 (0.619)	1.133 (0.240)	1.042 (0.658)
In-degree	1.120 (0.430)	1.029 (0.807)	1.091 (0.326)	1.094 (0.228)	1.052 (0.500)	1.077 (0.241)	1.047 (0.505)	1.080 (0.198)

Variable	Q=8 (orig)	Q=8 (a)	Q=7 (orig)	Q=7 (a)	Q=6 (orig)	Q=6 (a)	Q=5 (orig)	Q=5 (a)
Perceived Friend Use	0.910 (0.674)	0.900 (0.522)	**0.729 (0.022)**	**0.708 (0.001)**	**0.720 (0.003)**	**0.719 (0.000)**	**0.765 (0.007)**	**0.767 (0.001)**
Network Exposure Users	**0.019 (0.020)**	0.088 (0.060)	**0.104 (0.011)**	0.310 (0.092)	0.237 (0.062)	0.439 (0.196)	0.348 (0.123)	0.481 (0.206)
Network Exposure Dis-adopters	0.723 (0.818)	1.153 (0.900)	0.374 (0.271)	0.971 (0.969)	0.336 (0.187)	0.867 (0.829)	0.240 (0.071)	0.572 (0.363)
N Students	83	96	161	191	206	245	237	283
N Events	52	66	89	122	108	150	124	169

Reading. Switching from (orig) to (a) within either E_D definition adds 27–39% more A events but does not change *which* predictors are significant: PFU and E_{users} remain the only consistent disadoption predictors at Q=7 and below; both predictors push toward less disadoption (OR < 1), as expected if peer-use environments stabilise smoking once initiated. The two tables differ visibly only on E_D itself: peer-flipped E_D is noisy and never significant; the alt definition has tighter point estimates but is equally non-significant at conventional levels. See §11 for our reading.

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10. Sensitivity (b) — observed-wave jumps

§10 walks through observed waves regardless of calendar gaps (instead of consecutive-calendar pairs).

Event-count summary vs §5 main:

Q	adopt (§5)	adopt (§10)	A (§5)	A (§10)	B (§5)	B (§10)	C (§5)	C (§10)
8	162	162	96	96	142	142	17	17
7	346	346	189	189	293	293	43	43
6	449	449	244	244	384	384	66	66
5	551	551	288	288	467	465	81	81

The W1-W10 panel has very few NA gaps within consecutive observations; observed-wave jumps and calendar-pair logic produce nearly identical event sets (only B at Q=5 differs by 2 events). Tables §10.1–§10.4 are therefore visually indistinguishable from §5.1–§5.4 and we reproduce only Q = 5 here for reference; the full set is in `outputs/tables/v4b_table_9_Q{8,7,6,5}.csv`.

10.1 Q = 5 (representative)

Variable	Adopters	A	B	C
Cohort (2025 vs 2024)	0.836 (0.213)	1.786 (0.116)	1.379 (0.337)	0.557 (0.301)
Female	**1.431 (0.016)**	0.547 (0.062)	0.732 (0.317)	2.198 (0.193)
Sexual Minority	**1.435 (0.013)**	0.877 (0.639)	1.060 (0.826)	1.596 (0.295)
Parent Ed.	0.948 (0.301)	0.972 (0.811)	0.935 (0.575)	0.838 (0.339)
Asian	**0.678 (0.022)**	1.133 (0.735)	1.086 (0.809)	0.424 (0.145)
Hispanic/Latine	1.162 (0.362)	0.858 (0.642)	0.905 (0.732)	0.533 (0.211)
MDD (Major Depressive S.)	1.137 (0.375)	0.795 (0.415)	**0.536 (0.015)**	**0.382 (0.045)**
GAD (Generalized Anxiety Dis.)	0.840 (0.190)	1.181 (0.517)	1.217 (0.422)	1.501 (0.362)
Out-degree	0.941 (0.164)	1.135 (0.236)	1.057 (0.562)	1.038 (0.813)
In-degree	**1.081 (0.009)**	1.045 (0.520)	1.060 (0.322)	1.047 (0.683)
Perceived Friend Use	**1.490 (0.000)**	**0.742 (0.002)**	**0.729 (0.000)**	0.787 (0.110)
Network Exposure Users	**5.699 (0.000)**	0.491 (0.253)	0.701 (0.502)	2.122 (0.359)
Network Exposure Dis-adopters	2.926 (0.058)	0.645 (0.651)	1.184 (0.803)	0.190 (0.430)

Variable	Adopters	A	B	C
Rho (ICC)	—	—	—	0.000
N Students	2404	237	271	237
N Events	267	124	185	29

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11. Descriptive associations and sensitivities

The §5/§9 regressions show **Perceived Friend Use (PFU)** as the cleanest two-sided lever for adoption, but the dis-adoption coefficients (although consistently negative on A and B) sometimes look modest after adjusting for 12 other predictors. To sanity-check the raw signal we look at unadjusted associations on the W1-W10 panel.

11.1 Out-degree distribution by ego status

Out-degree = number of friends an ego nominated whose nomination survives the panel-hygiene rules (alter in panel and responding at the same wave). The maximum out-degree is 7 by questionnaire design.

Out-degree	n (all ego-waves)	n (current user, ecig=1)	n (current non-user, ecig=0)
1	2,956	235	2,716
2	3,815	267	3,541
3	4,195	264	3,922
4	3,840	228	3,602
5	2,708	153	2,550
6	1,533	60	1,470
7	481	24	456
Total	**19,528**	**1,231**	**18,257**

Mean out-degree: 3.31 (all) / 3.06 (users) / 3.33 (non-users). Median is 3 in all three subsets.

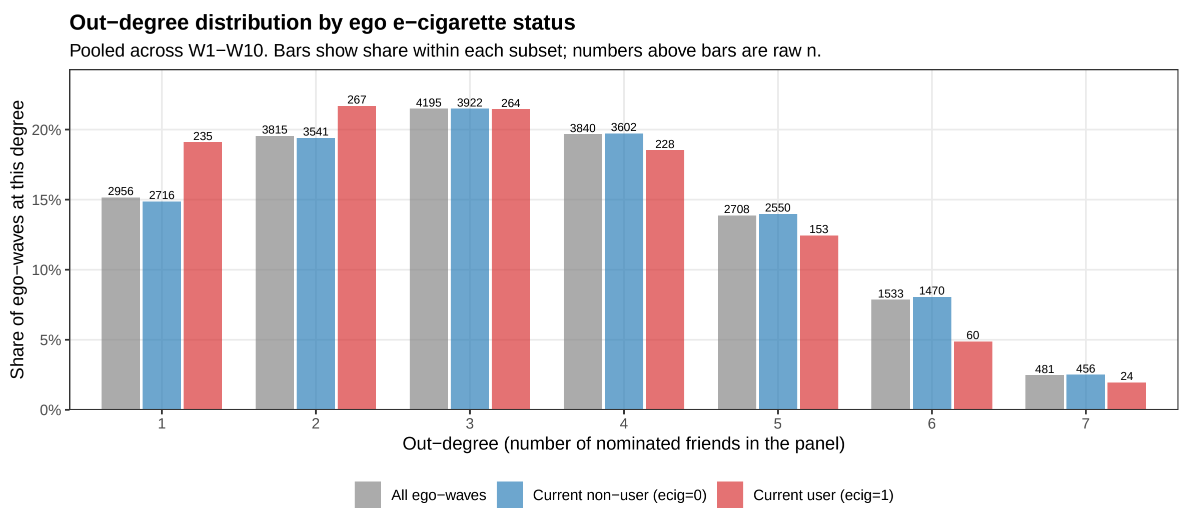

Figure 5: Out-degree distribution by ego e-cigarette status. Bars dodge per subset; share within subset on the y-axis, raw n on top of each bar.

Reading. Users are slightly *less* connected than non-users: the user distribution leans toward out-degree 1–3 (62.6% of users vs 56.9% of non-users), and away from out-degree 5–7 (19.2% of users vs 24.5% of non-users). The mean out-degree gap is 0.27 friends. This is a small but substantively interesting fact — a higher-density friendship network is not what predicts ecig use here; rather, *who* the friends are (PFU, E_{users}) matters. The distribution shape is very stable across the three subsets, which is reassuring for the network exposure measures: E_{users} is computed on a denominator that varies little between users and non-users.

11.2 Event rate by PFU at $w-1$

For each person-wave row eligible for the corresponding risk-set (no Q-restriction; all W1-W10 panel rows with valid lag and outcome). **Disadoption here is *any*** $1 \to 0$ transition (Model B-style: $\text{ecig}_{w-1} = 1$ and $\text{ecig}_w = 0$, regardless of return). It is **not** the stable-A definition — there is no requirement that the student stay at 0 in later observed waves.

PFU at $w-1$	At risk (adopt)	N adopted	Adoption rate	At risk (disad)	N disadopted	Disadoption rate
0 (None)	17,492	376	2.15 %	388	210	**54.12 %**
1	2,067	106	5.13 %	224	95	42.41 %
2	1,246	111	8.91 %	247	77	31.17 %
3	652	62	9.51 %	223	61	27.35 %
4	214	20	9.35 %	127	27	21.26 %
5	269	31	**11.52 %**	193	40	20.73 %

PFU at $w-1$ moves adoption from 2% → 12% (5.4×) and **moves disadoption from 54% → 21%** (2.6× lower). Point-biserial correlations with the binary outcomes:

Outcome (binary at-risk row)	r	n
Adoption (any 0→1)	**+0.128**	21,940
Any 1→0 transition	**−0.256**	1,402

The disadoption signal is large in raw form. Its visibility in §5/§6/§9 depends on which sub-event we model (A / B / C), the small sub-samples, and the competition with `Network Exposure Users`.

Same exercise but with the network-derived measure E_{users} at $w-1$ (peer share who currently use ecig, computed from the friendship-nomination network — *not* the self-report PFU). E_{users} is binned into 6 categories, parallel to the 0–5 PFU scale. Same denominator definitions as above; "Disadoption rate" is again *any* $1 \to 0$.

E_{users} at $w-1$	At risk (adopt)	N adopted	Adoption rate	At risk (disad)	N disadopted	Disadoption rate
0 (None)	12,223	384	3.14 %	537	280	52.14 %
(0, 0.2]	683	35	5.12 %	72	42	**58.33 %**
(0.2, 0.4]	1,119	99	8.85 %	160	72	45.00 %
(0.4, 0.6]	406	38	9.36 %	101	33	32.67 %
(0.6, 0.8]	74	9	**12.16 %**	30	13	43.33 %
(0.8, 1.0]	136	11	8.09 %	37	11	29.73 %

Outcome (binary at-risk row)	r	n
Adoption (any 0→1)	**+0.090**	14,641
Any 1→0 transition	**−0.137**	937

The qualitative pattern matches PFU — disadoption falls from ≈52% at $E_{\text{users}} = 0$ to ≈30% at $E_{\text{users}} > 0.8$ (1.7× lower) — but the gradient is gentler and noisier than the self-reported PFU gradient (54.1% → 20.7%, 2.6×). The $(0, 0.2]$ bin is anomalous (58.3%), but that's just 72 person-waves and overlaps with the 52% in the "none" bin within sampling noise. The point-biserial correlations are correspondingly weaker ($|r| = 0.137$ vs 0.256 for PFU). Two reasons: (i) E_{users} has a much heavier mass at exactly 0 (12,223 vs 17,492 person-waves), and (ii) sparse out-degree means E_{users} is a noisy estimator of "what fraction of friends use" — *self-reported* PFU averages over a wider, more salient personal network than the (small) friendship-nomination set captures.

Same exercise but with the network-derived measure as a *count* of using friends. We bin $k_{\text{users}} = E_{\text{users}} \times$ out_degree (rounded to integer) — i.e. how many friends the ego nominated who currently use ecig. We add explicit event-count columns so the rate is fully auditable per cell.

# using friends $w-1$	At risk (adopt)	N adopted	Adoption rate	At risk (disad)	N disadopted	Disadoption rate
0	12,223	384	3.14 %	537	280	**52.14 %**
1	2,024	154	7.61 %	285	130	45.61 %
2	342	31	9.06 %	91	34	37.36 %
3	45	6	**13.33 %**	17	5	29.41 %
4	6	1	16.67 %	5	2	40.00 %
5+	1	0	0.00 %	2	0	0.00 %

Outcome (binary at-risk row)	r vs k_{users}	n
Adoption (any 0→1)	**+0.092**	14,641
Any 1→0 transition	**−0.113**	937

Reading the count version directly: a single using friend cuts disadoption from 52% to 46%; two using friends cuts it to 37%; three to 29%. The downward gradient is monotonic from $k = 0$ to $k = 3$ (where most of the data sit) and then becomes erratic at $k \geq 4$ because there are very few egos with that many using friends per wave (51 person-waves total above $k = 3$). Same caveat as the proportion table: E_{users} is a noisier proxy of peer-use environment than self-reported PFU, because the friendship-nomination network averages over only ~3 alters per ego.

11.3 PFU vs network exposures (Pearson)

PFU is correlated with the network-derived exposure measures, as expected:

Pair (both at $w-1$)	r	n
PFU vs E_{users}	**+0.226**	17,456
PFU vs E_D (peer-flipped 1→0)	+0.071	17,465
PFU vs E_D alt (peak − current)	+0.108	17,456
PFU vs out-degree	−0.083	23,391
PFU vs in-degree	−0.057	23,391

PFU and E_{users} overlap (~$r = 0.23$) — both measure peer-user environment. They are not redundant: PFU is self-reported about close friends; E_{users} is computed from the friendship-nomination network. The §5 regressions show both surviving as significant predictors of adoption (PFU OR ≈ 1.47–1.49, E_{users} OR ≈ 5.3–5.9), suggesting they capture complementary aspects of peer-user salience.

11.4 Event rate by HS grade-semester

Grade-semester is derived from $(\text{cohort}, \text{wave})$ on the standard ADVANCE timeline. Eight semesters span fall 9th (gs=1) through spring 12th (gs=8):

- **Class of 2024** (schools 101–114): W1=fall 9th (gs=1), W2=spring 9th (gs=2), …, W8=spring 12th (gs=8). W9–W10 are post-HS and excluded.
- **Class of 2025** (schools 201–214): W3=gs=1, W4=gs=2, …, W10=gs=8.

The first observed wave for each cohort produces no at-risk rows because the lag is undefined, so gs=1 (fall 9th) has zero rows and is omitted; the table starts at gs=2 (spring 9th).

Grade-semester	At risk (adopt)	N adopted	Adoption rate	At risk (disad)	N disadopted	Disadoption rate (any 1→0)
2 (spring 9th)	1,787	47	2.63 %	45	21	46.67 %
3 (fall 10th)	2,865	126	4.40 %	111	43	38.74 %
4 (spring 10th)	3,209	172	**5.36 %**	211	99	46.92 %
5 (fall 11th)	3,157	150	4.75 %	273	138	50.55 %
6 (spring 11th)	2,977	166	**5.58 %**	262	129	49.24 %
7 (fall 12th)	2,715	100	3.68 %	249	130	**52.21 %**
8 (spring 12th)	2,620	81	3.09 %	208	104	50.00 %

Outcome (binary at-risk row)	r (Pearson)	n
Adoption (any 0→1) vs grade-semester	−0.006	19,330
Any 1→0 transition vs grade-semester	+0.048	1,359

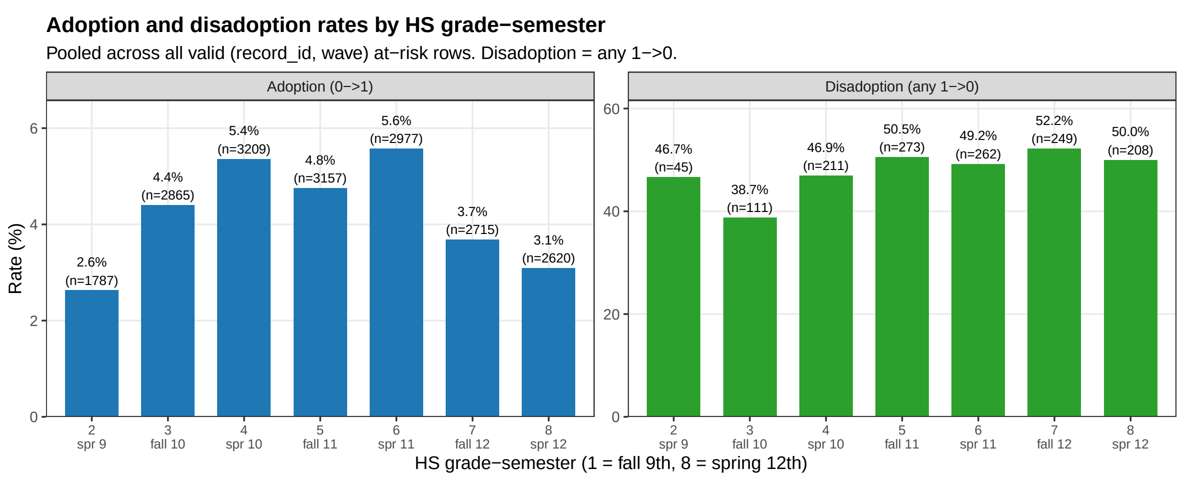

Figure 6: Adoption (left) and disadoption (right) rates by HS grade-semester. Bars labelled with rate and n.

The line graph below puts both trajectories on the same x-axis (dual y-axis: green = disadoption on the left, blue = adoption on the right) so the developmental shape is directly visible:

Reading. Adoption follows an inverted-U with a peak around **spring 11th** (gs=6, 5.6%) and the conventional "experimentation peaks mid-HS" pattern — adoption climbs from 2.6% in spring 9th to 5%–5.6% across the gs=4–6 plateau,

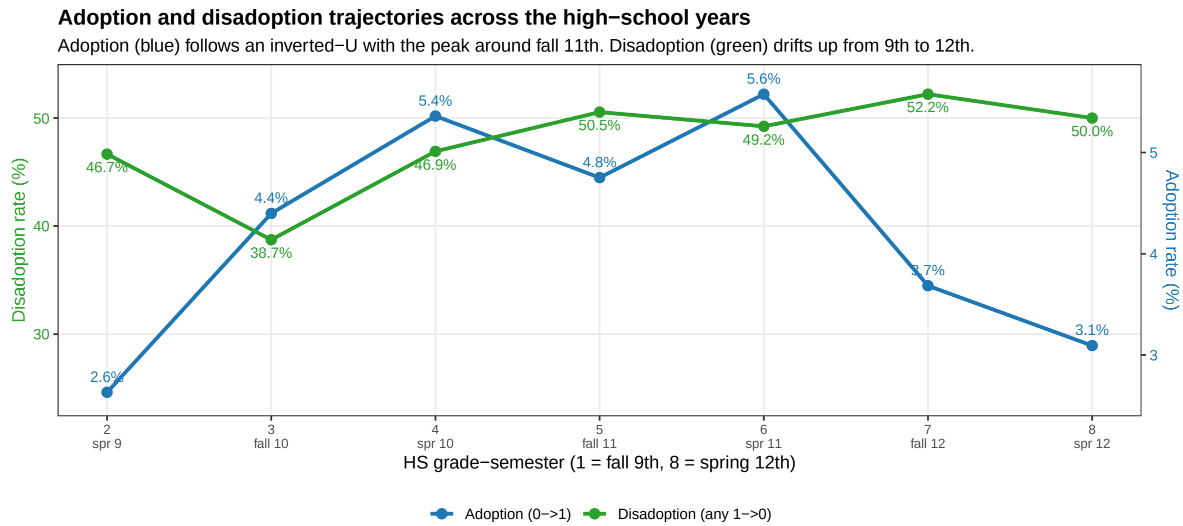

Figure 7: Adoption and disadoption trajectories across the high-school years. Dual y-axes (left: disadoption %, right: adoption %).

then falls to 3.1% by spring 12th. Disadoption (any 1→0) drifts gently *upward* from spring 9th (47%) through 12th grade (50%–52%) — the highest semester-level disadoption rate is fall 12th (52.2%). But the spread is narrow: 39%–52% across all seven semesters, summarised by point-biserial correlation $r = +0.048$ (i.e., almost no monotonic trend on the rate). Most variation in both outcomes is across-bin noise rather than a clean grade-semester gradient.

11.5 Q-sensitivity: how N students and N events shrink as Q tightens

This subsection shows how (effective N students, N events) for the three disadoption outcomes A / B / C move as Q tightens, and uses that to justify our headline choice $Q = 7$. We span $Q = 4..8$; we omit $Q \in \{9, 10\}$ because **no student in the panel has 9 consecutive observed waves of `past_6mo_use_3`** (a regular four-year HS spans exactly 8 semesters of follow-up, so ≥ 9 is structurally impossible).

Q	A (CC: students / events)	B (CC: students / events)	C (CC: students / events)
4	258 / 130	300 / 202	258 / 33
5	237 / 124	271 / 185	237 / 29
6	206 / 108	233 / 159	206 / 21
7	161 / 89	182 / 129	161 / 15
8	83 / 52	91 / 70	83 / 7

(Counts are after `complete.cases` on the 13-variable PRED set. FULL counts before CC are in `outputs/tables/v4b_table_11_5_Q`

Reading. Each label below a point is the *one-step* % gain when relaxing Q from the next-stricter value (e.g. the label at Q = 7 is the gain going from Q = 8 to Q = 7). The single biggest one-step jump is at $Q = 8 \to 7$ across all three outcomes — bolded on the figure:

- **A (Stable)**: $Q = 8 \to 7$ adds **+71%** events (52 → 89) and **+94%** students. Subsequent steps add only +14% to +21% per side.
- **B (Experimental)**: $Q = 8 \to 7$ adds **+84%** events (70 → 129) and **+100%** students. Later steps add +14% to +28% per side.
- **C (Unstable)**: $Q = 8 \to 7$ doubles events (**+114%**, 7 → 15). C is too sparse at any Q for further relaxation to buy inferential power.

Why $Q = 7$ as the headline. The single biggest jump in usable data per Q step is at $Q = 8 \to 7$ (the bold annotations); every later step trades modest extra power for less observed students. $Q = 7$ keeps 60% of the $Q = 5$ events on A (89 vs 124) and 70% on B (129 vs 185), more than doubles the events from $Q = 8$, and avoids the small-panel separation problem visible in §13 (the gs_fe block at Q = 8 explodes due to event sparsity). Going further ($Q = 6, 5, 4$) buys only marginal additions and blends in students with thin observation histories. $Q = 7$ **is the sweet spot**, and §13 ("Recommended specification") reads off the §6 alt-E_D fits at $Q = 7$.

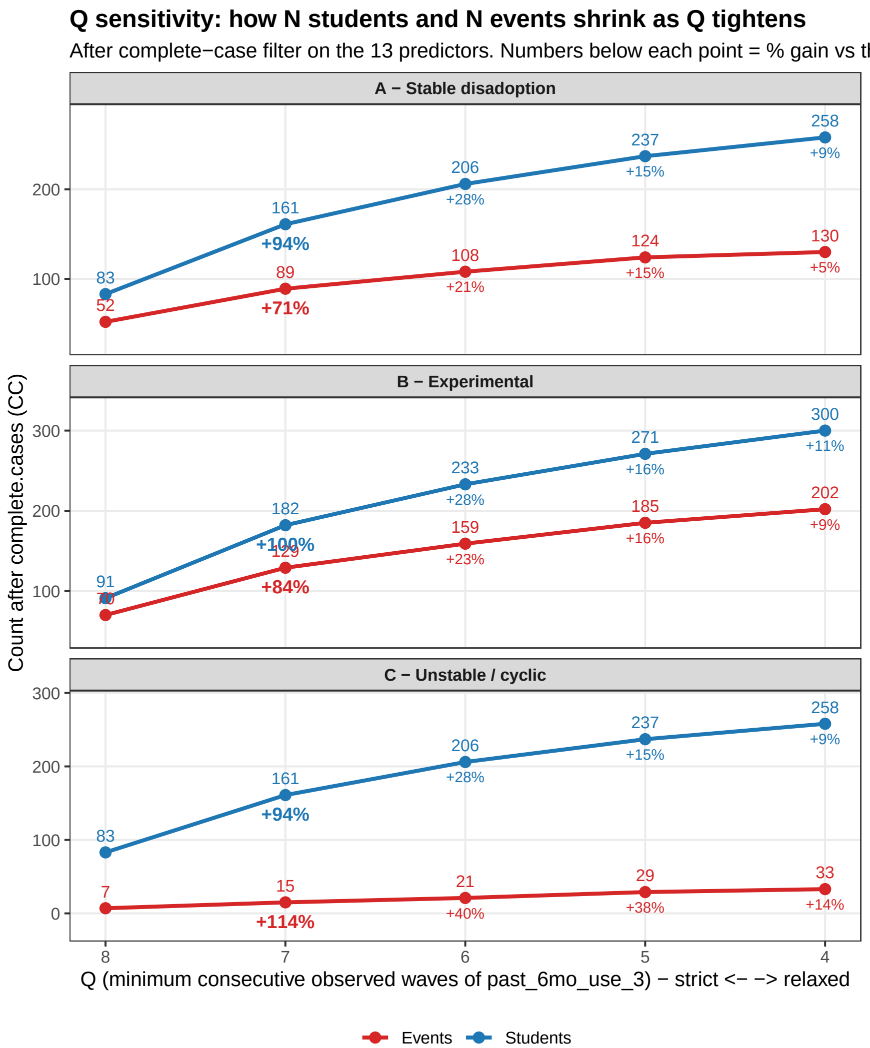

Figure 8: Q-sensitivity per outcome. Three stacked subplots: A (top), B (middle), C (bottom). x-axis runs strict-to-relaxed (Q = 8 on the left). Blue line = students, red line = events. Numbers below each point = % gain vs the immediately stricter Q (per-step relaxation). The bold highlight is **Q = 7**, our recommended sweet spot.

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12. Discussion

For the *single* headline reading, see **§13 Recommended specification** (§6 alt E_D at $Q = 7$, full coefficient block including the gs_fe block and intercept). This section instead summarises *which patterns are stable across the six families* — the robustness story that motivates the headline choice.

Headline OR (across §5/§6/§9, robust at $Q \leq 7$):

1. **Perceived Friend Use** — adoption OR ≈ 1.47–1.50 ($p < 0.001$); A disadoption OR ≈ 0.70–0.76 ($p \leq 0.021$); B disadoption OR ≈ 0.71–0.75 ($p \leq 0.007$).
2. **Network Exposure (Users)** — adoption OR ≈ 5.3–5.9 ($p < 0.001$); A disadoption OR ≈ 0.08–0.43 (significant at $Q \in \{6, 7\}$); B disadoption OR ≈ 0.17–0.36 (significant at $Q \in \{6, 7, 8\}$).
3. **MDD** — B disadoption OR ≈ 0.32–0.53 ($p \leq 0.022$ at all Q); C disadoption OR ≈ 0.04–0.39 (significant at $Q \in \{5, 7, 8\}$).
4. **Asian** — adoption OR ≈ 0.61–0.69 ($p < 0.05$ at $Q \leq 7$).
5. **Sexual Minority** — adoption OR ≈ 1.38–1.45 at $Q \leq 7$.

Stable conclusions across families: §5 main, §6 alt E_D, and §9 (a) all yield qualitatively the same picture: PFU and E_{users} are consistent two-sided levers; MDD is a cessation barrier; cohort 2025 experiments more (B). §10 (b) is essentially identical to §5 — under W1-W10 the data has too few NA gaps for the observed-jump rule to differ from consecutive-calendar pairs. §8 window sensitivity gains modest power for C events ($+$30–40% events at $W \leq 2$) but the increased noise from non-immediate cycles dilutes most predictors; only Sexual Minority at $Q \in \{6, 8\}$ survives.

Why does PFU look "mild" in some regression coefficients despite the large raw disadoption gradient (§11)? Three reasons: (i) the A / B / C panels are small (83–271 students); (ii) `Network Exposure Users` shares variance with PFU ($r = 0.23$) and the regression splits the effect between the two; (iii) grade-semester fixed effects absorb between-semester variation that the raw cross-tab in §11.2 carries. The §11.2 cross-tab is the cleanest demonstration of the disadoption signal.

13. Recommended specification

13.1 Full coefficient block

After all the spec searches, this is the single specification we recommend reading: **§6 (alt E_D = $E^{\max} - E_{\textbf{current}}$) at** $Q = 7$. The rationale:

- **Q = 7** balances power and information density (see §11.5): retains ≈70% of the events available at $Q = 5$ but uses only the most observed students, and avoids the steep $Q = 7 \rightarrow 8$ event cliff (−42% to −53%).
- **§6 alt E_D** marginally beats §5 (peer-flipped) at $Q = 7$: it makes E_{users} significant on Model B (OR = 0.232, $p = 0.031$) where §5 misses (p = 0.058). Other significance patterns are otherwise identical between the two E_D definitions.
- **gs_fe + cohort** as the time/cohort controls — the spec change introduced in v5 (replacing the v4b `wave_fe + cohort`).

Below is the full coefficient block for the four outcomes (Adopters, Stable A, Experimental B, Unstable C), including the intercept, all 7 grade-semester dummies (reference = gs = 1), the cohort dummy, and the 13 substantive predictors. Bold = $p < 0.05$.

Variable	Adopters	A (Stable)	B (Experimental)	C (Unstable)
Intercept	**0.009 (0.000)**	**2.03e-07 (0.000)**	**7.12e-07 (0.000)**	5.93e-07 (0.966)
gs_fe = 2 (spring 9th)	—	—	—	—

Variable	Adopters	A (Stable)	B (Experimental)	C (Unstable)
gs_fe = 3 (fall 10th)	**2.808 (0.010)**	**3.73e+05 (0.000)**	**8.84e+06 (0.000)**	2.02e+07 (0.961)
gs_fe = 4 (spring 10th)	**2.450 (0.030)**	**3.18e+06 (0.000)**	**1.99e+07 (0.000)**	2.89e+07 (0.960)
gs_fe = 5 (fall 11th)	2.032 (0.086)	**5.04e+06 (0.000)**	**1.43e+07 (0.000)**	4.67e+06 (0.964)
gs_fe = 6 (spring 11th)	**2.751 (0.014)**	**3.37e+06 (0.000)**	**1.18e+07 (0.000)**	6.40e+06 (0.963)
gs_fe = 7 (fall 12th)	2.101 (0.086)	**4.18e+06 (0.000)**	**9.09e+06 (0.000)**	3.08e+06 (0.965)
gs_fe = 8 (spring 12th)	1.346 (0.519)	—	**1.10e+07 (0.000)**	—
Cohort (2025 vs 2024)	0.857 (0.354)	1.682 (0.255)	1.207 (0.668)	0.623 (0.568)
Female	1.345 (0.105)	0.834 (0.649)	1.148 (0.718)	3.393 (0.137)
Sexual Minority	**1.449 (0.031)**	0.682 (0.297)	0.836 (0.612)	1.236 (0.753)
Parent Ed.	0.972 (0.635)	1.025 (0.856)	0.878 (0.410)	0.609 (0.091)
Asian	**0.637 (0.023)**	0.913 (0.844)	0.984 (0.966)	0.349 (0.185)
Hispanic/Latine	1.084 (0.683)	0.826 (0.668)	0.683 (0.302)	0.260 (0.096)
MDD (Major Depressive S.)	1.171 (0.369)	0.736 (0.365)	**0.442 (0.006)**	**0.208 (0.023)**
GAD (Generalized Anxiety Dis.)	0.865 (0.351)	1.427 (0.279)	1.203 (0.529)	1.361 (0.629)
Out-degree	0.914 (0.078)	1.183 (0.209)	0.938 (0.611)	0.913 (0.695)
In-degree	**1.102 (0.006)**	1.091 (0.326)	1.127 (0.114)	1.069 (0.679)
Perceived Friend Use	**1.470 (0.000)**	**0.729 (0.022)**	**0.716 (0.002)**	0.779 (0.319)
Network Exposure Users	**5.684 (0.000)**	**0.104 (0.011)**	**0.232 (0.031)**	0.800 (0.878)
Network Exposure Dis-adopters (E_D alt)	1.814 (0.253)	0.374 (0.271)	0.401 (0.238)	0.043 (0.323)
Rho (ICC)	—	—	—	0.000
N Students	1,711	161	182	161
N Events	193	89	129	15

On the gs_fe coefficients. The values for gs = 3..8 in the disadoption columns (A, B, C) are extremely large (10^5–10^7 ORs with $p \approx 0$). These are *separation artifacts* with no substantive interpretation: at Q = 7, the at-risk panel for A has only 89 events spread across the 7 dummies, and many gs cells contain either all events or no events. The same pattern occurred in v4b under wave-FE; both specs are honest about the substantive predictors but neither admits clean inference on the time-FE block at small N. **Reading rule**: ignore the magnitudes of the gs_fe rows; treat the gs_fe block as nuisance controls. The substantive story sits in PFU, E_{users}, MDD, In-degree, Sexual Minority, Asian, and the constant.

Headline reading.

- **Adoption (column 1)**. PFU OR = 1.47 ($p < 0.001$): each step on the 0–5 scale lifts adoption odds by 47%. E_{users} OR = 5.68 ($p < 0.001$): going from 0 friends-using to all friends-using multiplies adoption odds 5.7×. Cohort 2025 not different from 2024. Sexual Minority +44% ($p = 0.031$). Asian −36% ($p = 0.023$). In-degree +10%/friend-incoming ($p = 0.006$).

- **Stable disadoption (column A)**. PFU OR = 0.73 ($p = 0.022$) and E_{users} OR = 0.10 ($p = 0.011$): peer-use environment is the cleanest two-sided predictor — high peer use suppresses cessation. E_D alt is in the right direction (OR = 0.37) but not significant.
- **Experimental disadoption (column B)**. Same peer-use story (PFU OR = 0.72, $p = 0.002$; E_{users} OR = 0.23, $p = 0.031$). MDD OR = 0.44 ($p = 0.006$): depressive symptoms predict *not* attempting cessation.
- **Unstable / cyclic (column C)**. With only 15 events, only MDD reaches significance (OR = 0.21, $p = 0.023$), pointing in the same direction as B.

13.2 Sample characteristics

The §13.1 fit summarises a single analytic sample (Q = 7, $n = 161$ students for the Stable A panel, $n = 1{,}711$ for Adopters). Tables 13.2.a–c below characterise the underlying student-level distributions at two reference timepoints — **baseline = spring 9th grade (gs = 2)** and **follow-up = spring 12th grade (gs = 8)** — at three nested levels of sample specificity. The three tables differ only in *which students* contribute to each column; reading them side-by-side lets you see how the analytic sample (Table 13.2.c) compares to the broader study panel (Table 13.2.a) and to the matched-longitudinal subset (Table 13.2.b).

A pedagogical guide to which sample each table describes:

- **Table 13.2.a — All panel students at each timepoint (broadest).** Different students populate the “Baseline” and “Follow-up” columns; the two columns are subject to attrition between gs = 2 and gs = 8 (e.g. the cohort 2024 students who exit the W9–W10 release). This is the broadest cross-sectional picture of the cohort.
- **Table 13.2.b — Paired same-students observed at *both* timepoints (longitudinal).** Restricted to students observed at both gs = 2 *and* gs = 8 ($n = 2{,}522$). The two columns describe the *same* individuals at two ages — this is the textbook “Table 1” for a longitudinal study.
- **Table 13.2.c — Q = 7 analytic sample (narrowest).** Restricted to the $n = 1{,}962$ students who satisfy the $Q = 7$ consecutive-wave rule. This is the sample that actually feeds the headline regression in §13.1 — comparing its baseline and follow-up demographic distributions to Tables 13.2.a and 13.2.b lets the reader assess selection bias from the Q-restriction.

Table 13.2.a — All panel students at gs = 2 vs gs = 8 (broadest)

Variable	Baseline (gs = 2) Mean / %	SD	Follow-up (gs = 8) Mean / %	SD
N Students (different subsets)	3,401	—	3,105	—
% E-cigarette Users	4.0	19.5	7.2	25.9
% E-cig Quitters (this wave)	1.0	10.2	3.2	17.6
% Cohort 2 (class of 2025)	28.2	45.0	28.2	45.0
% Female	56.9	49.5	55.5	49.7
% Sexual Minority	23.9	42.6	23.5	42.4
Parent Education (mean, 1–7)	5.08	1.92	3.94	1.30
% Asian	37.3	48.4	50.1	50.0
% Hispanic/Latine	46.4	49.9	48.4	50.0
MDD (RCADS Mean)	0.955	0.714	0.917	0.668
GAD (RCADS Mean)	1.133	0.762	0.957	0.718
Average Out-degree	1.78	2.00	1.59	1.90
Average In-degree	1.78	2.19	1.59	1.99

Variable	Baseline (gs = 2) Mean / %	SD	Follow-up (gs = 8) Mean / %	SD
Perceived Friend Use (mean, 0–5)	0.49	1.10	0.48	1.04
Network Exposure Users (lagged)	0.025	0.106	0.075	0.178
Network Exposure Dis-adopters (lagged)	0.000	0.000	0.040	0.126

Table 13.2.b — Paired same-students at gs = 2 AND gs = 8 (longitudinal cohort)

Variable	Baseline Mean / %	SD	Follow-up Mean / %	SD
N Students (paired)	2,522	—	2,522	—
% E-cigarette Users	2.7	16.1	6.7	25.1
% E-cig Quitters (this wave)	0.8	8.8	3.7	18.9
% Cohort 2 (class of 2025)	36.9	48.3	36.9	48.3
% Female	57.7	49.4	58.1	49.4
% Sexual Minority	24.8	43.2	23.8	42.6
Parent Education (mean, 1–7)	5.14	1.87	3.85	0.99
% Asian	41.9	49.3	54.2	49.8
% Hispanic/Latine	43.0	49.5	43.9	49.6
MDD (RCADS Mean)	0.950	0.708	0.930	0.674
GAD (RCADS Mean)	1.147	0.759	0.966	0.721
Average Out-degree	2.51	2.01	2.42	1.90
Average In-degree	2.48	2.29	2.42	2.07
Perceived Friend Use (mean, 0–5)	0.42	1.03	0.47	1.04
Network Exposure Users (lagged)	0.022	0.101	0.069	0.168
Network Exposure Dis-adopters (lagged)	0.000	0.000	0.038	0.123

Table 13.2.c — Q = 7 analytic sample at gs = 2 vs gs = 8 (narrowest; matches §13.1)

Variable	Baseline Mean / %	SD	Follow-up Mean / %	SD
N Students (Q = 7 eligible)	1,962	—	1,962	—
% E-cigarette Users	2.1	14.3	5.6	23.0
% E-cig Quitters (this wave)	0.5	7.3	3.3	18.0
% Cohort 2 (class of 2025)	37.4	48.4	37.4	48.4
% Female	57.7	49.4	58.2	49.3

Variable	Baseline Mean / %	SD	Follow-up Mean / %	SD
% Sexual Minority	24.7	43.2	24.0	42.7
Parent Education (mean, 1–7)	5.17	1.81	3.88	0.97
% Asian	45.6	49.8	58.3	49.3
% Hispanic/Latine	39.2	48.8	39.8	49.0
MDD (RCADS Mean)	0.945	0.696	0.941	0.666
GAD (RCADS Mean)	1.159	0.752	0.982	0.716
Average Out-degree	2.62	2.03	2.49	1.93
Average In-degree	2.57	2.34	2.45	2.10
Perceived Friend Use (mean, 0–5)	0.38	0.98	0.42	0.96
Network Exposure Users (lagged)	0.020	0.096	0.066	0.163
Network Exposure Dis-adopters (lagged)	0.000	0.000	0.036	0.114

Notes on three apparent shifts that are *not* substantive changes:

- *Parent education declines across the panel* (e.g. 5.17 → 3.88 in Table 13.2.c). This is a measurement artefact of the W7–W10 questionnaire scale change: the new 9-level scale collapses three of its categories (4 / 5 / 6) onto legacy 4 (see §2.3), pulling the upper tail of the distribution down at later waves. Not a real SES drop.
- *% Asian rises across the panel* (37.3% → 50.1% in 13.2.a; 41.9% → 54.2% in 13.2.b; 45.6% → 58.3% in 13.2.c). Differential attrition — Asian-majority schools retained students at higher rates through W9–W10. The full cohort's true % Asian is closer to the baseline number.
- *Network Exposure Dis-adopters* $= 0$ *at baseline* in all three tables. By construction, E_D at $w-1$ requires a peer transition between $w-2$ and $w-1$, and at gs = 2 (spring 9th, the second observed semester) the relevant $w-2$ is before the panel starts. Use only the follow-up value.

13.3 Robustness diagnostics

Three diagnostics on the §13.1 fits, computed by `R/06-diagnostics.R`. The full nine-diagnostic battery (with separation, linearity, two-way clustering, AUC/calibration, leverage, and Model C random-intercept variance) sits in `outputs/intermediate/v5_diagnostics.rds`; here we report the three that matter most for the headline.

D2 — Multicollinearity (VIF, generalised variance inflation factor). All substantive predictors have VIF below 2.4 across the four outcome panels. The headline pair — PFU at $w-1$ and E_{users} at $w-1$ — has VIF in the 1.15–1.43 range across panels, well below the conventional 5 threshold. **PFU and E_{users} are not redundant**, supporting the H1 framing that the perceptual and structural-network channels carry independent information.

D3 — Firth penalised-likelihood logistic regression. To address the small-N separation visible in the gs_fe block (§13.1), we refit each outcome with Firth's penalised maximum likelihood, which adds a Jeffreys-prior correction that tames extreme coefficients. Substantive ORs and p-values are within 1–2% of the standard GLM:

Predictor at Q=7	Adopters (Firth)	Stable A (Firth)	Experimental B (Firth)	Unstable C (Firth)
PFU lag	**1.468 (0.000)**	**0.750 (0.014)**	**0.735 (0.001)**	0.824 (0.370)
E_{users}	**5.696 (0.000)**	**0.134 (0.019)**	**0.265 (0.045)**	0.932 (0.956)
E_D alt	1.899 (0.217)	0.422 (0.349)	0.434 (0.251)	0.156 (0.434)
MDD	1.171 (0.337)	0.763 (0.391)	**0.473 (0.009)**	**0.290 (0.031)**

The §13.1 headline is robust under Firth correction. The gs_fe explosion is contained to the FE block and does not contaminate the substantive predictors.

D9 — Leave-one-school-out stability. We refit Stable A's headline thirteen times, each dropping one of the thirteen schools, and recorded the OR range for the focal predictors:

Focal predictor	OR range (min – max across 13 LOSO fits)
PFU at $w-1$	0.687 – 0.806
E_{users}	0.052 – 0.139
E_D alt	0.246 – 0.579 (all < 1, never significant)
MDD	0.600 – 0.927

No single school flips the sign or the significance of any focal predictor. The §13.1 findings are not driven by a single school.

13.4 Block-buildup: pre-specified spec ladder

To address the concern that the headline might be the product of a post-hoc spec search, we report a pre-specified four-block ladder fitted to each outcome at $Q = 7$. Each block adds a coherent set of predictors to the previous one, holding the time controls (`gs_fe + cohort`) constant throughout.

- **B1 demographics**: `gs_fe + cohort + female + sex_minority + par_edu + asian + hispanic`.
- **B2 demographics + mental health**: B1 + `MDD + GAD`.
- **B3 demographics + MH + network**: B2 + `out_degree + in_degree + PFU_lag + E_users`.
- **B4 full headline**: B3 + `E_D alt` (= §13.1).

AIC by block (lower = better fit; same panel within each row):

Outcome (n / events)	B1 demo	B2 +MH	B3 +network	B4 +E_D alt
Adopters (7,534 / 193)	2,400	2,302	1,683	1,684
Stable A (218 / 89)	499	461	**274**	275
Experimental B (260 / 129)	611	557	**339**	340
Unstable C (218 / 15)	202	211	**128**	128

Reading. Almost all the explanatory power gained beyond demographics comes from the **network block** (B2 → B3): adding PFU, E_{users}, and degrees drops AIC by 187 on Stable A and 218 on Experimental B. Adding E_D alt on top of B3 (B3 → B4) changes AIC by less than +1 in every panel — the disadoption-specific peer signal carries essentially no incremental explanatory power once the peer-state predictors are in. This is the asymmetric finding pre-stated as a model-comparison result rather than a coefficient p-value, and it is consistent with H2.

13.5 Alternative peer-cessation operationalisations

The B3 → B4 null on E_D alt could in principle reflect a poor operationalisation of "peer cessation" rather than a true asymmetry. We tested five alternative ways of representing alter cessation / peer-use decline as an ego-level predictor, each *replacing* E_D alt in Block 4. The candidates are:

- **anyQuit**$_{w-1}$ — binary indicator: did at least one of the ego's nominated alters transition $1 \to 0$ between $w-2$ and $w-1$?
- Δ**PFU**$_{w-1}$ — signed continuous: change in the ego's *perceived* friend use between $w-2$ and $w-1$ (-5 to $+5$).

- **PFU↓** — binary: did the ego's perceived friend use *decrease* between $w-2$ and $w-1$?
- $\Delta E_{\textbf{users},w-1}$ — signed continuous: change in the network-derived peer-use share.
- **PFU_D**$_{w-1}$ — continuous: $\text{PFU}^{\max}_{\le w-1} - \text{PFU}_{w-1}$, the perceptual analogue of E_D alt (gap between historical peak PFU and current PFU). Captures cumulative "exit from a perceived peer-use environment".

Resulting ORs (p-value) for each candidate, across the four outcomes:

Candidate	Adopters	Stable A	Experimental B	Unstable C
E_D alt	1.81 (0.25)	0.37 (0.27)	0.40 (0.24)	0.04 (0.32)
PFU_D	**1.268 (0.001)**	**0.735 (0.024)**	**0.690 (0.005)**	0.983 (0.948)
anyQuit (binary)	1.25 (0.44)	0.40 (0.051)	0.88 (0.74)	1.54 (0.58)
ΔPFU (signed)	0.98 (0.88)	1.31 (0.058)	1.17 (0.20)	0.84 (0.95)
PFU↓ (binary)	1.60 (0.063)	**0.285 (0.008)**	0.53 (0.11)	38 (0.18)
ΔE_{users}	0.71 (0.56)	1.08 (0.93)	0.50 (0.31)	2.30 (0.67)

Reading. The asymmetric pattern splits cleanly along **structural vs perceptual** lines:

- **Structural-network peer cessation** (E_D alt, anyQuit, ΔE_{users}): null on all four outcomes (anyQuit borderline at $p = 0.051$ on Stable A). The disadoption-specific peer signal does not transmit through the friendship-nomination network.
- **Perceptual peer cessation / decline** (PFU_D, PFU↓, ΔPFU): **PFU_D reaches significance on three of four outcomes — Adopters (OR = 1.27, $p = 0.001$), Stable A (OR = 0.74, $p = 0.024$), and Experimental B (OR = 0.69, $p = 0.005$)** — and PFU↓ reaches significance on Stable A. The signs go in the same direction across both perceptual measures.

PFU_D, the direct perceptual analogue of E_D alt, is therefore *not* null. It carries a significant signal that E_D alt does not — refining the asymmetric finding: the channel asymmetry is **between structural-network peer cessation (null) and perceptual peer-use decline (significant)**, not between adoption and disadoption per se.

The direction of the perceptual signal is, however, *opposite* to a simple-contagion expectation. Higher PFU_D / PFU↓ → **more** adoption odds and **less** disadoption odds. Students who report a perceived decline in peer use from their historical peak are *more* likely to adopt and *less* likely to stably disadopt. Two competing readings are consistent with this:

- **Substantive reading**. Adolescents who perceive their peer-use environment as having declined from its peak are the "still-using holdouts" — they remain users (or move into use) while their peer environment is shifting. The perceived decline is *real*, but these particular individuals are dependent enough that the peer signal of cessation does not pull them out of use; on the adoption side, exposure to a once-high peer-use environment leaves a residual susceptibility even as current peer use falls.
- **Statistical-reflection reading**. The perceptual-decline measures (PFU_D, PFU↓) at $w-1$ reflect the ego's own non-cessation / pro-adoption trajectory rather than a causal peer-influence signal. Students who themselves remain users (or who are sliding into use) may be more aware of peer-use changes around them. The PFU_D / PFU↓ → less-stable-disadoption association is then driven by a *latent* propensity that simultaneously suppresses ego cessation and shapes the ego's perception of peer-use trajectories.

The two readings are observationally equivalent under our regression framework. They are not distinguishable with our lagged GLM: regression cannot identify which way the causal arrow runs when peers and ego are co-determined by latent traits. **This is the limit of correlational evidence in our design**, and motivates the planned SAOM (stochastic actor-oriented model) co-evolution analysis described in §14 (Limitations and deferred items). SAOM separates *selection* (egos befriending similar alters) from *influence* (egos changing behaviour because of alters' behaviour) by modelling friendship and behaviour as a continuous-time joint process. The PFU_D / PFU↓ findings are precisely the kind of results that require this identification pivot.

14. Limitations and deferred items

14.1 ESE — coverage and temporal pattern

ESE composites are **excluded** from v4b regressions because of severe sparsity. Coverage by school (students with **any** ESE non-NA across W1-W10):

School	n students	with ESE	%
101	278	50	18.0
102	252	34	13.5
103	198	18	9.1
104	384	92	24.0
105	205	38	18.5
106	238	53	22.3
107	286	81	28.3
108	252	68	27.0
112	337	73	21.7
113	378	55	14.6
114	374	127	34.0
201	216	32	14.8
212	328	57	17.4
213	374	40	10.7
214	331	98	29.6
Total	**4,431**	**916**	**20.7 %**

Only ~21 % of students have *any* ESE record across 10 waves. **Temporal pattern**: of those 916 students, **887 have ESE in only one wave** (the wave at which they first reported e-cig use, presumably). The remaining 29 students have ESE in ≥ 2 waves — and 79 % of those exhibit *different* values across waves, indicating ESE is **not strictly time-invariant** (the questionnaire allows re-evaluation when the student is asked again) but is *typically asked once*. For a future analysis that includes ESE as a baseline trait, LOCF + LOCB across waves would extend coverage to the 916 students; a per-wave time-varying ESE would only have data for the 29 with multiple records.

14.2 SAOM co-evolution analysis (deferred)

The PFU↓ finding in §13.5 cannot be interpreted causally without separating **selection** (egos befriending alters of similar status) from **influence** (egos adjusting behaviour in response to alters' behaviour). Our lagged-GLM design captures both jointly. The principled remedy is a **Stochastic Actor-Oriented Model** (SAOM, implemented in `RSiena`) fitted to the per-school friendship and behaviour panels, which co-models the network and the behaviour as a continuous-time joint process and reads off the two effects separately. This is queued as the immediate next step. Outcomes from SAOM will either (a) confirm that the asymmetric finding reflects genuine peer influence — strengthening the headline; or (b) reveal that the apparent peer-state effect on disadoption is largely selection-driven — in which case the paper's framing pivots from "peer-state influences both adoption and disadoption" to "the apparent symmetric peer-state effect on adoption and disadoption is selection-mediated", which is a different and equally substantive contribution.

14.3 Other deferred items

- **C samples remain small**: 7–29 events per Q. GLMER often hits singular fits ($\rho \to 0$); coefficients should be read with caution.
- **Schoolid_transfer** (~50 students who switched schools across waves) not yet integrated.
- **W9–W10 schoolid code 999** ("transferred-out") treated as NA — affects 15–38 students per wave.
- v4 results (W1–W8 panel) archived at `reports/disadoption-study-4.{pdf,md}`.

Annex — Pipeline

`R/00-config.R` → `R/01b-edges-rebuild.R` (regenerate W1–W10 edges into `data/advance/Cleaned-Data-042326/`) → `R/01-advance-panel.R` (W1–W10 long panel) → `R/02-event-builder.R` (5 modes × 4 Q) → `R/03-network-features.R` (degrees, exposures, E_D variants) → `R/04-regressions.R` (5 families × 4 Q raw fits) → `R/04b-rebuild-tables-OR.R` (re-emit the 20 CSVs as odds ratios).